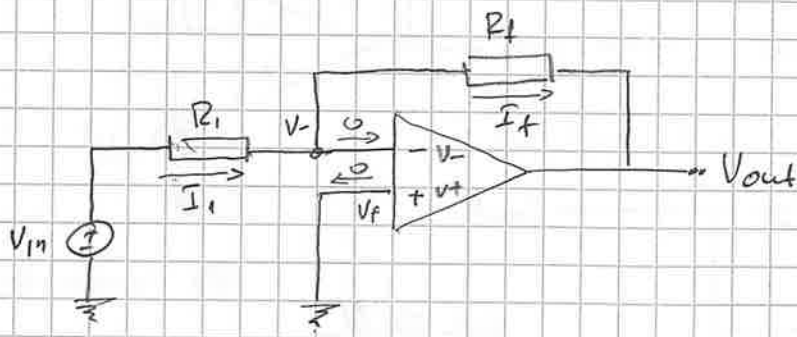


OP - INVERTER



$$I_- = 0$$

$$I_+ = 0$$

$$V_+ = 0$$

$$V_- = 0$$

$$V_{out} = A(V_{in} - 0)$$

↓
ground

$$A = \frac{V_{out}}{V_{in}}$$

KCL V_-

$$I_1 = I_f$$

$$\frac{V_{in} - 0}{R_1} = \frac{0 - V_o}{R_f} \Rightarrow \frac{V_{in}}{R_1} = - \frac{V_o}{R_f}$$

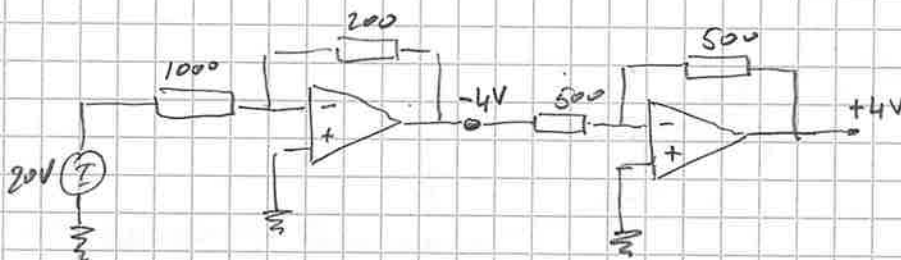
$$\frac{R_f}{R_1} V_{in} = -V_o$$

$$V_o = - \frac{R_f}{R_1} V_{in}$$

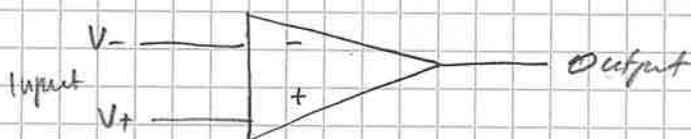
if: $R_f = R_1 \Rightarrow V_o = -V_{in}$

$R_f = 10R_1 \Rightarrow V_o = -10V_{in}$

$R_f = \frac{1}{3}R_1 \Rightarrow V_o = -\frac{1}{3}V_{in}$



OP description = Ideal OP



$$V_o = A(V_- - V_+)$$

$$10,000 - (0.0001) = 1$$

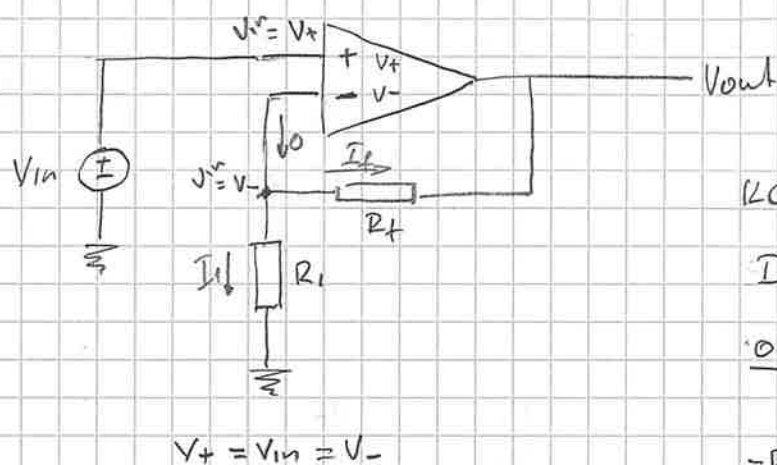
$\frac{V_{out}}{V_{in}} = A$

1) No current ever flows into either inputs terminals

2) There is NO voltage difference between the two input terminals.

$$V_- = V_+$$

OP - NOT INVERTER



$$V_+ = V_{in} = V_-$$

if: $R_f = 0 \rightarrow V_{out} = V_{in}$

$R_f = R_1 \rightarrow V_{out} = 2 V_{in}$

$R_f = 10 R_1 \rightarrow V_{out} = 11 V_{in}$

$$\left(1 + \frac{R_f}{R_1}\right) = \left(1 + \frac{10 R_1}{R_1}\right) = (1 + 10) = 11$$

$R_f = \frac{1}{10} R_1 \rightarrow V_{out} = 1.1 V_{in}$

KCL @ V_-

$$I_1 = I_f$$

$$\frac{0 - V_{in}}{R_1} = \frac{V_{in} - V_{out}}{R_f}$$

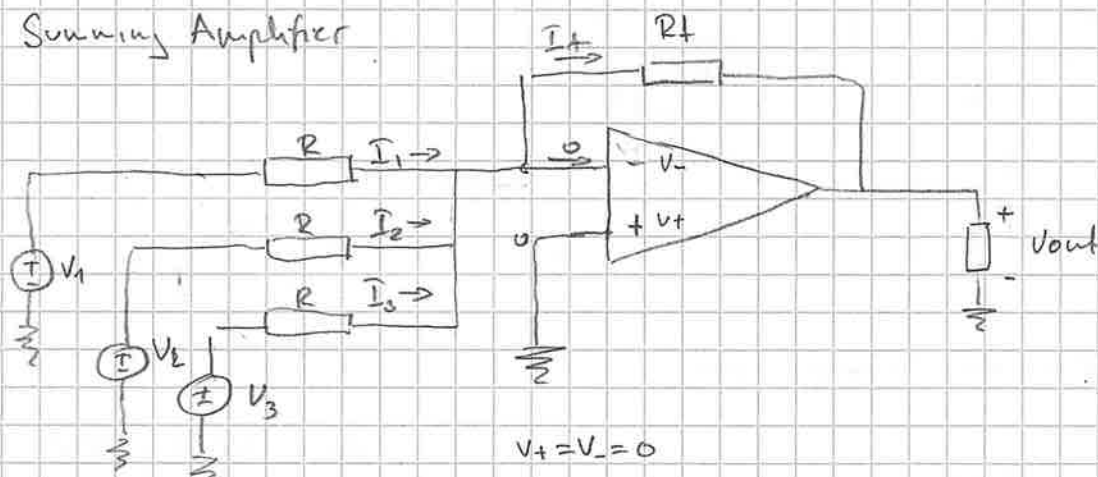
$$-R_f V_{in} = R_f V_{in} - R_f V_{out}$$

$$R_f V_{out} = (R_1 + R_f) V_{in}$$

$$V_{out} = \left(\frac{R_1 + R_f}{R_1}\right) V_{in}$$

$$V_{out} = \left(1 + \frac{R_f}{R_1}\right) V_{in}$$

Summing Amplifier

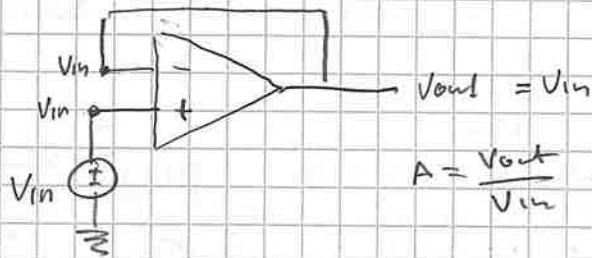


KCL @ V_-

$$I_1 + I_2 + I_3 = I_f$$

$$\left(\frac{V_1 - 0}{R} + \frac{V_2 - 0}{R} + \frac{V_3 - 0}{R}\right) = \frac{0 - V_{out}}{R_f}$$

$$R_f (V_1 + V_2 + V_3) = -R V_{out} \Rightarrow -\frac{R_f}{R_1} (V_1 + V_2 + V_3) = V_{out}$$

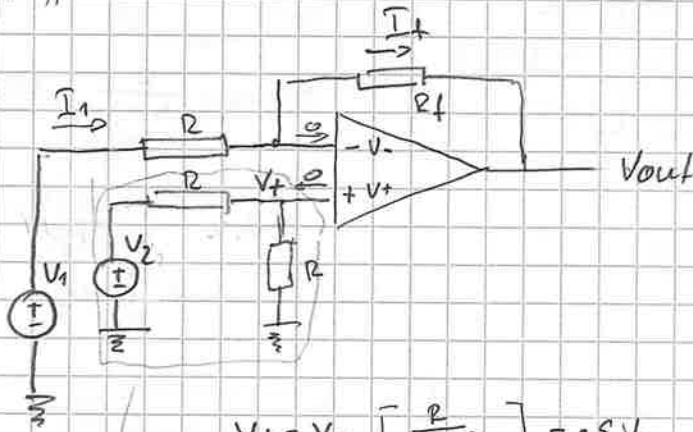


$$A = \frac{V_{out}}{V_{in}} = 1$$

Voltage follower / Buffer amplifier

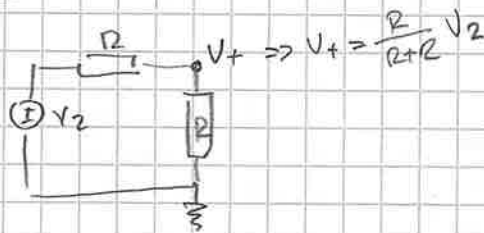
- * high input impedance
- * low output impedance
- * Voltage gain = 1

Difference OP



$$V_+ = V_2 \left[\frac{R}{R+R} \right] = 0.5 V_2$$

$$V_- = V_+ = 0.5 V_2$$



$$I_1 = I_f$$

$$\frac{V_1 - (V_-)}{R} = \frac{(V_-) - V_{out}}{R_f}$$

$$\frac{V_1 - 0.5 V_2}{R} = \frac{0.5 V_2 - V_{out}}{R_f}$$

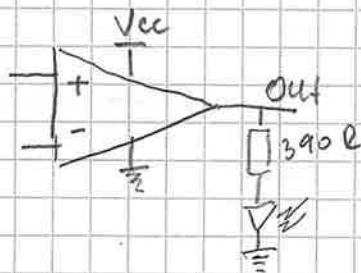
$$V_1 R_f - 0.5 V_2 R_f = 0.5 V_2 R - V_{out} R$$

$$\frac{R_f V_1 - \frac{1}{2} V_2 (R_f + R)}{-R} = \frac{-R V_{out}}{-R}$$

$$\frac{1}{2} V_2 \left(\frac{R_f + R}{R} \right) - \frac{R_f}{R} V_1 = V_{out}$$

$$\left(\frac{R_f + R}{2R} \right) V_2 - \left(\frac{R_f}{R} \right) V_1 = V_{out}$$

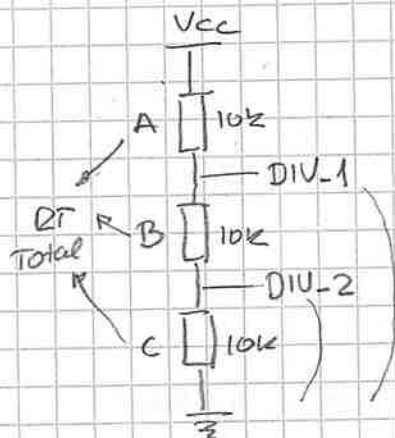
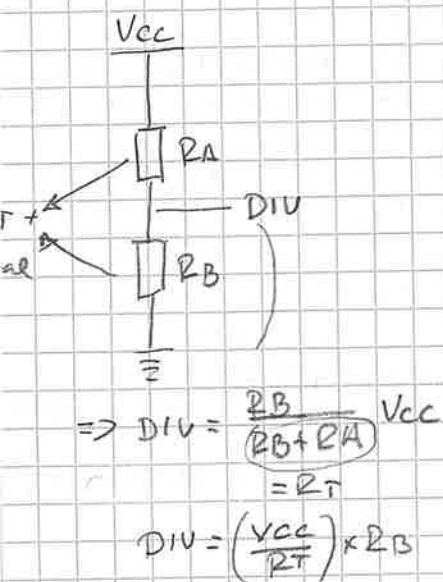
OP Comparator



$$V_+ > V_- \Rightarrow V_{out} = V_{cc}$$

+	-	Out	
2V	1V	5V	$V_+ > V_-$
3V	2V	5V	$V_+ > V_-$
4.5	4.6	0V	$V_+ < V_-$
1.1	1.2	0V	$V_+ < V_-$

Voltage divider



$$\text{DIV-1} = \frac{B+C}{A+B+C} V_{CC}$$

or

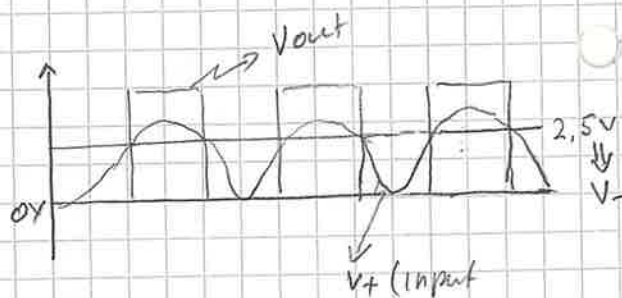
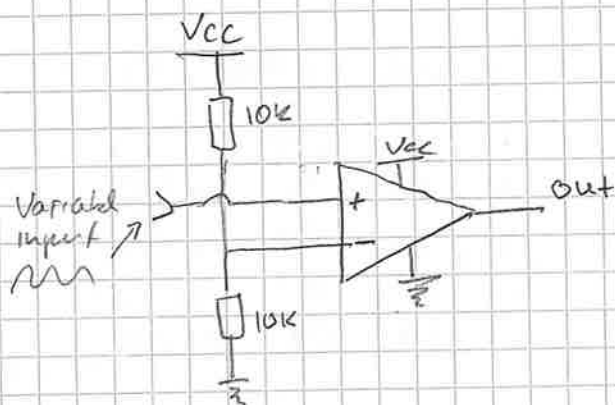
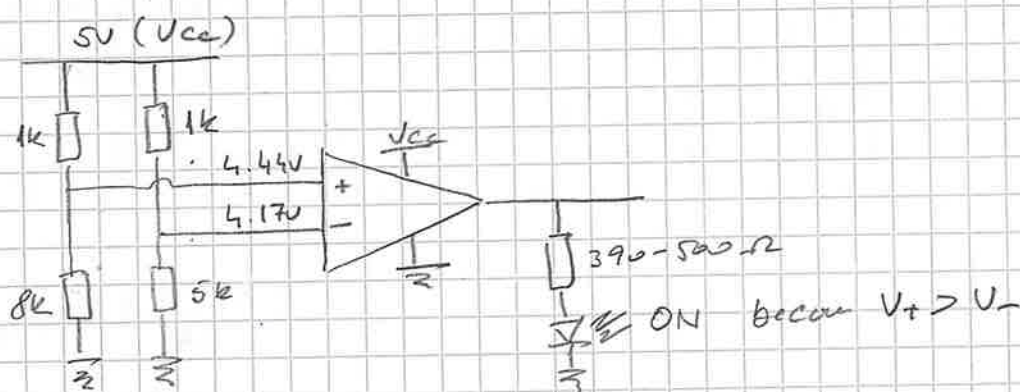
$$\text{DIV-1} = \left(\frac{V_{CC}}{R_T} \right) \times (B+C)$$

$$\text{DIV-2} = \frac{C}{A+B+C} V_{CC}$$

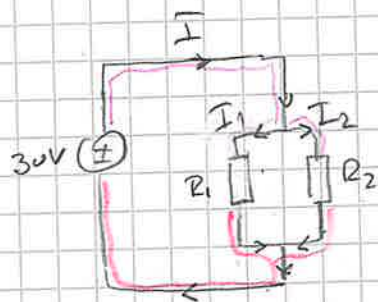
or

$$\text{DIV-2} = \left(\frac{V_{CC}}{R_T} \right) \times C$$

Example: OP Comparator + Voltage divider



Parallel Circuit - Current



$$R_1 = 150\Omega$$

$$R_2 = 100\Omega$$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}$$

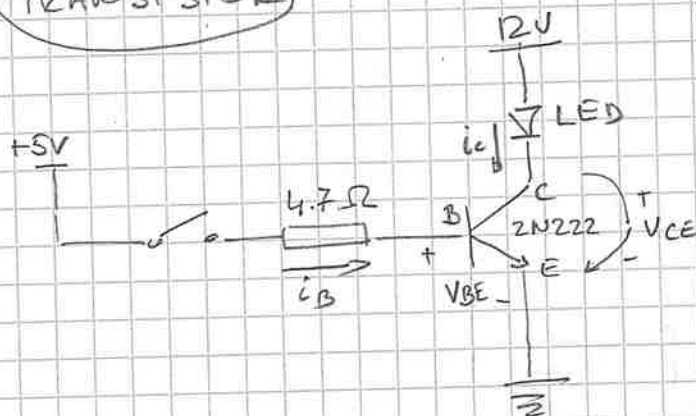
$$\Rightarrow R_{eq} = \left(\frac{1}{150\Omega} + \frac{1}{100\Omega} \right)^{-1} = 60\Omega$$

$$I = \frac{30V}{R_{eq}} = \frac{30V}{60\Omega} = 0.5A$$

$$I_1 = \frac{30V}{R_1} = \frac{30V}{150\Omega} = 0.2A$$

$$I_2 = \frac{30V}{R_2} = \frac{30V}{100\Omega} = 0.3A$$

TRANSISTOR NPN



$$12V - V_{LED} - V_{CE} = 0 \Rightarrow V_{CE} = 12V - V_{LED}$$

$$V_{CE_max} = 1V \text{ (datasheet)}$$

$$V_{LED_min} = 12V - V_{CE_max} = 11V$$

$$V_{LED} = 9V \text{ to } 14.8V$$

(with this range is 11V)

$$V_{BE_min} = 0.6V \text{ (datasheet)}$$

$$5V - I_B(4.7k\Omega) - V_{BE} = 0$$

$\Rightarrow$

$$I_B = \frac{5V - 0.6V}{4.7k\Omega} = 0.94mA$$

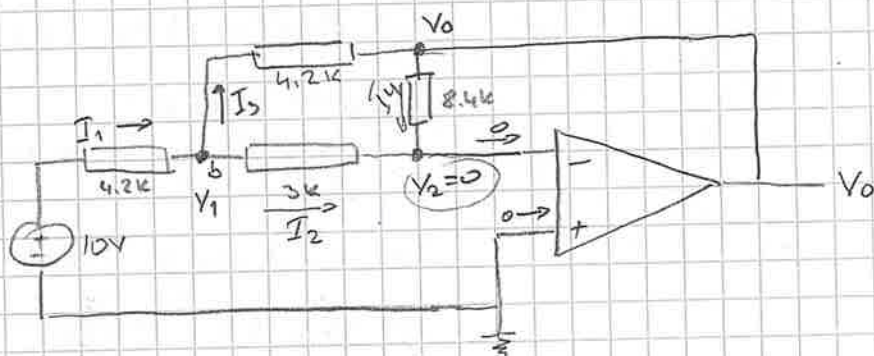
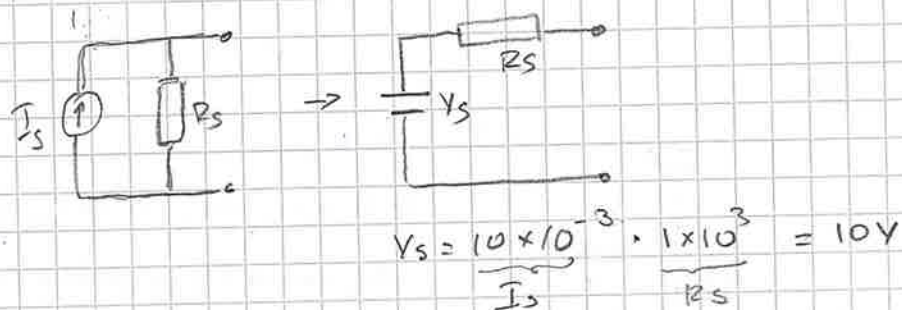
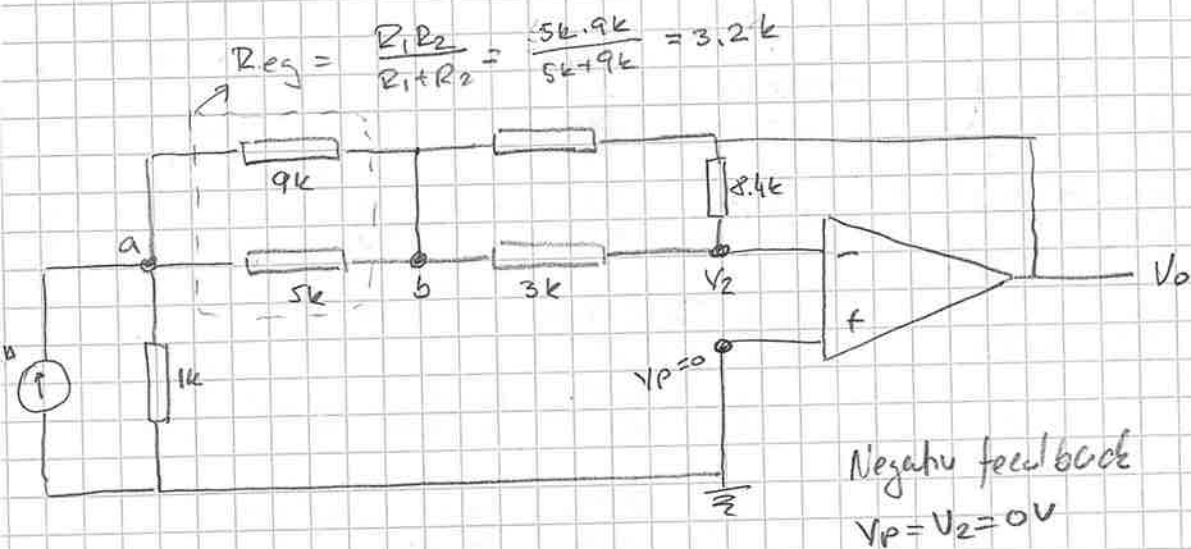
$$I_B = 0.94mA < I_{Arduino\ 10mA\ max} \checkmark$$

$$I_C = h_{FE_min} \cdot I_B = 35 \cdot I_B$$

datasheet

$$\Rightarrow I_C = 32.8mA > I_{LED}$$

Op Example:



Applying KCL at node - b

$$I_1 = I_2 + I_3$$

$$\frac{10 - V_1}{4.2k} = \frac{V_1 - V_2}{3k} + \frac{V_1 - V_o}{4.2k}$$

$$30 - 3kV_1 - 4.2kV_1 + 4.2kV_2 - 3V_1 + 3V_o = 0 \Rightarrow \frac{V_1}{3k} = -\frac{V_o}{8.4k}$$

$$30 - 10.2kV_1 + 3kV_o = 0$$

$$-10.2kV_1 + 3V_o = -30 \quad (I)$$

Applying KCL at inverting input

$$I_2 = I_4$$

$$\Rightarrow \frac{V_1 - V_2}{3k} = \frac{V_2 - V_o}{8.4k}$$

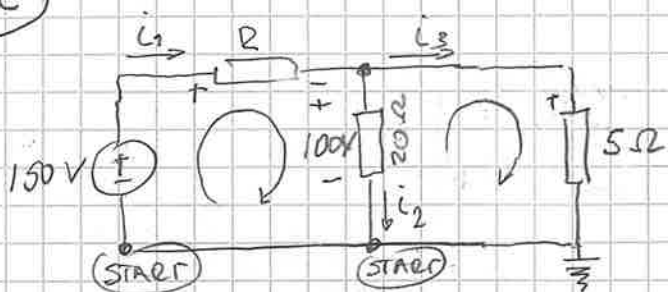
$$\Rightarrow 8.4kV_1 + 3V_o = 0 \quad (II)$$

Solving I and II

$$V_1 = 1.612V$$

$$V_2 = -4.516V$$

KCL



(1*) KCL @ Top Node

$$-i_1 + i_2 + i_3 = 0$$



$$V = i \cdot R$$

$$100 = i_2 \cdot 20$$

$$i_2 = 5A$$

KVL @ Left loop

$$-150V + i_1 R + 100V = 0$$

KVL @ Right loop

$$-100 + i_3 R = 0$$

$$5i_3 = 100$$

$$i_3 = 20A$$

$$\Rightarrow -i_1 + i_2 + i_3 = 0$$

$$-i_1 + 5A + 20A = 0$$

$$\Rightarrow i_1 = 25A$$

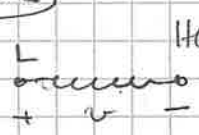
(2*)

$$-150V + i_1 R + 100V = 0$$

$$-150V + 25A R + 100V = 0$$

$$[R = 2\Omega]$$

INDUCTOR



Henry (H)

$$V = L \cdot \frac{di}{dt}$$

$$\tau = L/R$$

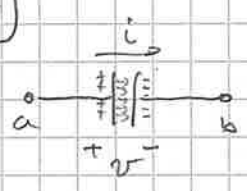
How fast current "i" changes thru inductor L

Note: Constant current $\Rightarrow 0V$ across L

Current Rises:

$\Rightarrow$ Positive voltage

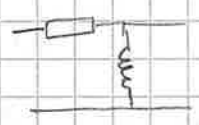
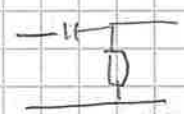
Capacitor



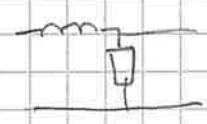
$$\tau = R \cdot C$$

$$i = C \frac{dv}{dt}$$

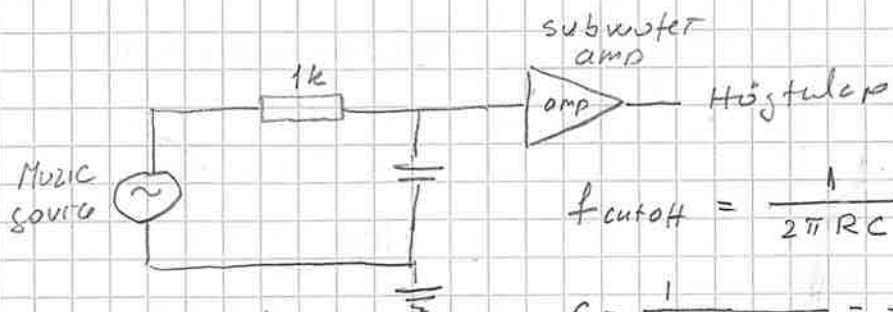
HP filter



LP filter



LOW PASS FILTER



$$f_{cutoff} = \frac{1}{2\pi RC} = 150 \text{ Hz}$$

$$C = \frac{1}{2\pi R f_{cutoff}} = \frac{1}{2\pi \cdot 1000 \cdot 150} = 1.06 \mu\text{F} \approx 1 \mu\text{F}$$

signal
 $f < f_{cutoff}$
 Input signal



filter output



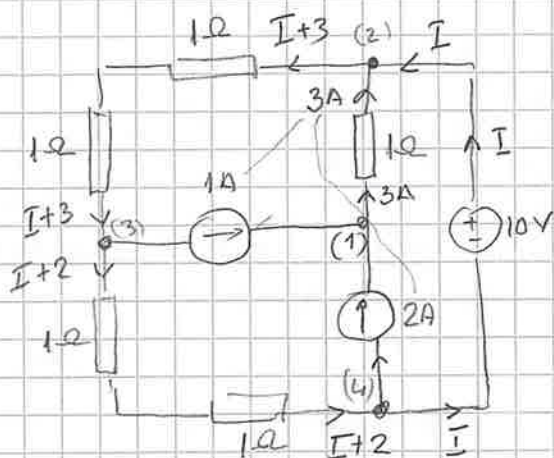
signal
 $f > f_{cutoff}$
 input signal



filter output



KCL



$$(1) \quad 1\text{A} + 2\text{A} = 3\text{A}$$

$$(2) \quad -I + (I+3) + 3 = 0$$

$$(3) \quad -(I+3) + (I+2) + 1 = 0$$

$$(4) \quad -(I+2) + 2 + I = 0$$

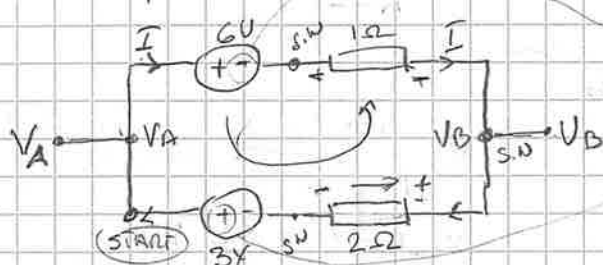
$$-10 - (I+3) - (I+3) - (I+2) - (I+2) = 0$$

$$10 - 10 - 4I = 0$$

$$I = 0$$

$$P_{del} = (10\text{V}) \times I = 10\text{V} \cdot 0 = 0$$

Exempel



$$\begin{aligned} + \rightarrow - &= - \\ - \rightarrow + &= + \end{aligned}$$

$$\begin{aligned} 6 - I - 2I + 3 &= 0 \\ \Rightarrow I &= -1 \end{aligned}$$

$$\begin{aligned} (1) \quad (V_A - V_B) - 6 - I &= 0 \\ \text{From } V_A \text{ to } V_B \end{aligned}$$

$$V_A - 6 - I = V_B$$

$$(V_A - V_B) = 6 + I$$

$$(V_A - V_B) = 6 - 1 = 5$$

$$(V_A - V_B) - 6 - I - 2I + 3 = 0$$

$$V_A - 6 - 3I + 3 = V_B$$

$$-6 - 3I + 3 = 0 \Rightarrow I = -1$$

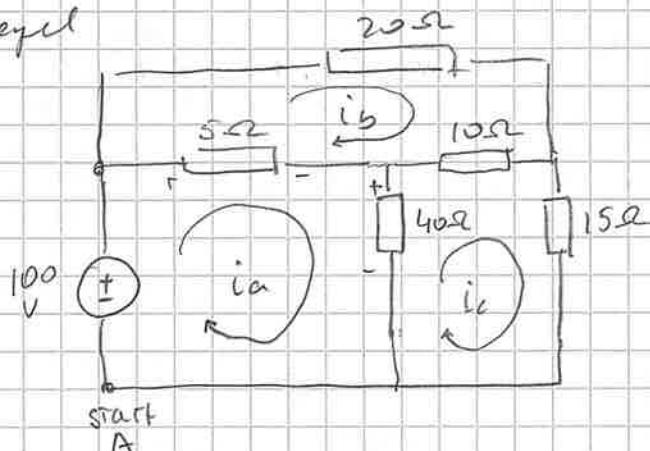
or (2)

$$V_A - 3 - 2I = V_B = 0$$

$$(V_A - V_B) - 3 + 2I = 0$$

$$(V_A - V_B) = 3 + 2 = 5$$

Exempel



$$\begin{aligned} A: -100 + (i_a - i_b)5 + (i_a - i_c)40 &= 0 \\ -100 + 5i_a - 5i_b + 40i_a - 40i_c &= 0 \\ * 45i_a - 5i_b - 40i_c &= 100 \end{aligned}$$

$$\begin{aligned} B: (i_b - i_a)5 + 20i_b + (i_b - i_c)10 &= 0 \\ * 5i_a + 35i_b - 10i_c &= 0 \end{aligned}$$

$$\begin{aligned} C: (i_c - i_a)40 + (i_c - i_b)10 + i_c15 &= 0 \\ 40i_c - 40i_a + 10i_c - 10i_b + 15i_c &= 0 \\ * -40i_a - 10i_b + 65i_c &= 0 \end{aligned}$$

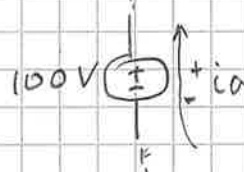
$$A \times B$$

$$\begin{bmatrix} 45 & -5 & -40 \\ -5 & 35 & -10 \\ -40 & -10 & 65 \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} = \begin{bmatrix} 100 \\ 0 \\ 0 \end{bmatrix}$$

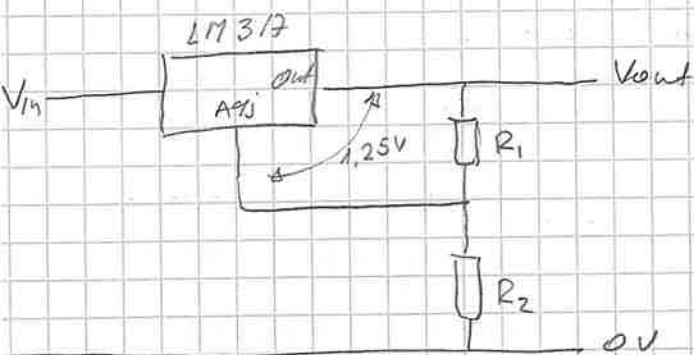
$$\begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} = A^{-1}B \begin{bmatrix} 100 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.06 & 0.02 & 0.04 \\ 0.02 & 0.037 & 0.018 \\ 0.04 & 0.018 & 0.043 \end{bmatrix} \begin{bmatrix} 100 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \\ 4 \end{bmatrix} \Rightarrow \begin{aligned} i_a &= 6 \\ i_b &= 2 \\ i_c &= 4 \end{aligned}$$

$$\begin{aligned} \text{Power source} \\ P &= -iV = -6 \cdot 100 = -600W \end{aligned}$$



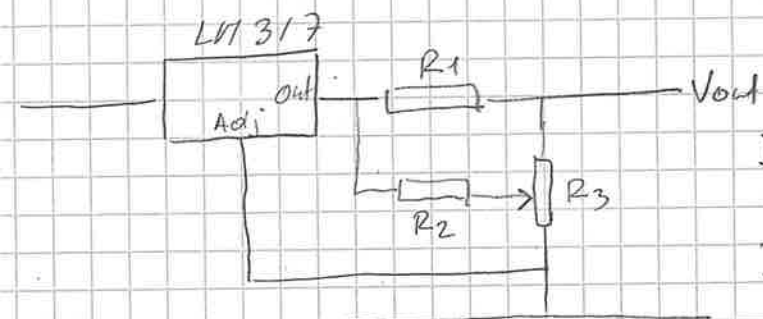
Linear regulator



$$V_{R1} = \frac{R_1}{R_1 + R_2} V_{out}$$

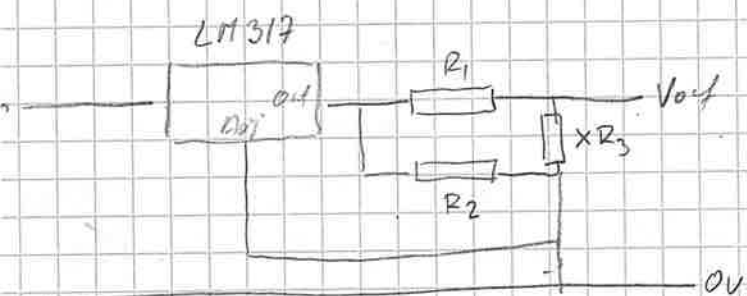
$$V_{out} = \left(1 + \frac{R_2}{R_1}\right) \left[\frac{V_{R1}}{1} = 1.25V\right]$$

$$V_{out} = 1.25 \left(1 + \frac{R_2}{R_1}\right)$$



$$I_{min} = \frac{1.25}{R_1}$$

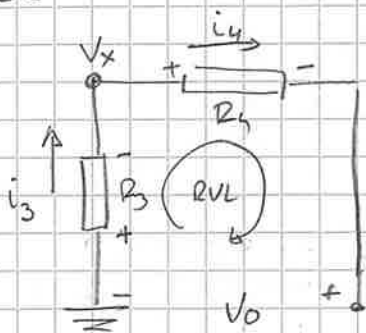
$$I_{max} = \left(1 + \frac{R_2}{R_1}\right) \frac{1.25}{R_1}$$



$$V_{R1} = \left(1 + \frac{XR_3}{R_2}\right) \frac{1.25}{0.000...1}$$

$$I_{R1} = \frac{1.25}{R_1} \left(1 + \frac{XR_3}{R_2}\right)$$

KCL

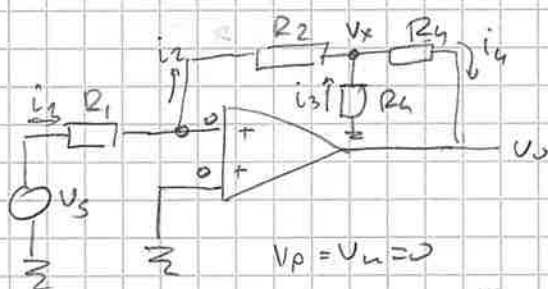


$$\text{KVL: } -i_3 R_3 - i_4 R_4 - V_o = 0 \quad \xrightarrow{+ \rightarrow -} = -$$

$$\Rightarrow -R_3 \left(\frac{R_2}{R_3 R_1} V_s \right) - R_4 \left(\frac{R_2}{R_1} \times V_s \left(\frac{1}{R_2} + \frac{1}{R_3} \right) \right) = V_o$$

$$\Rightarrow -V_s \left[\frac{R_2}{R_1} + \frac{R_4}{R_1} + \frac{R_4 R_2}{R_3 R_1} \right] = V_o$$

$$\Rightarrow \frac{V_o}{V_s} = \left[\frac{R_2}{R_1} + \frac{R_4}{R_1} + \frac{R_4 R_2}{R_3 R_1} \right] = A$$



$$V_p = V_n = 0$$

$$i_3 = \frac{0 - V_x}{R_3} = \frac{R_2}{R_1 R_3} V_s \quad \text{II}$$

$$i_2 = -\frac{1}{R_2} \times \left(-\frac{R_2}{R_1} V_s \right) = \frac{V_s}{R_1} \quad \text{III}$$

$$i_2 = \frac{V_n - V_x}{R_2} = -\frac{V_x}{R_2}$$

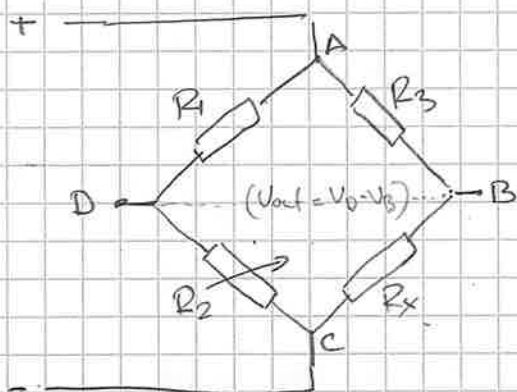
$$i_1 = i_2 \Rightarrow \frac{V_s}{R_1} = -\frac{V_x}{R_2}$$

$$V_x = -\frac{R_2}{R_1} V_s \quad \text{I}$$

$$V_x \text{ node: } i_2 + i_3 = i_4 \Rightarrow \frac{V_s}{R_1} + \frac{R_2}{R_1 R_3} V_s = i_4$$

$$\Rightarrow \frac{R_2}{R_1} V_s \left(\frac{1}{R_2} + \frac{1}{R_3} \right) = i_4 \quad \text{IV}$$

Wheatstone bridge



$$\frac{R_2}{R_1} = \frac{R_x}{R_3} \Rightarrow R_x = \frac{R_2}{R_1} R_3$$

$$\frac{V_{DC}}{V_{AD}} = \frac{V_{BC}}{V_{AB}}$$

$$\Rightarrow \frac{I_2 R_2}{I_1 R_1} = \frac{I_x R_x}{I_3 R_3}$$

$$R_D = \frac{R_2}{R_1 + R_2} \quad R_B = \frac{R_x}{R_3 + R_x}$$

$$V_{out} = V_D - V_B = V_{R_2} - V_{R_x} = 0$$

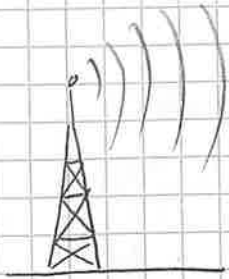
$$V_B = \frac{R_x}{R_3 + R_x} \times V_s$$

$$V_D = \frac{R_2}{R_1 + R_2} \times V_s$$

$$\text{Balance } R_D = R_B$$

$$\Rightarrow R_x = R_4 = \frac{R_2 R_3}{R_1}$$

anten

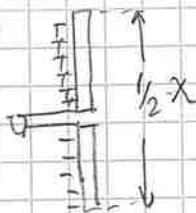


$$\lambda_0 = \frac{c}{f} = \frac{3 \cdot 10^8}{\frac{24 \cdot 10^8}{2.4 \text{ GHz}}} = 16.67 \text{ cm} \quad \text{in vacuum}$$

$$\lambda_g = \frac{c}{f \sqrt{\epsilon_{eff}}} = \frac{3 \cdot 10^8}{2.4 \text{ GHz} \sqrt{4.2}} \approx 6.1 \text{ cm}$$

PCB Layer

FR4



$$L \geq \frac{1}{20} \lambda_g$$

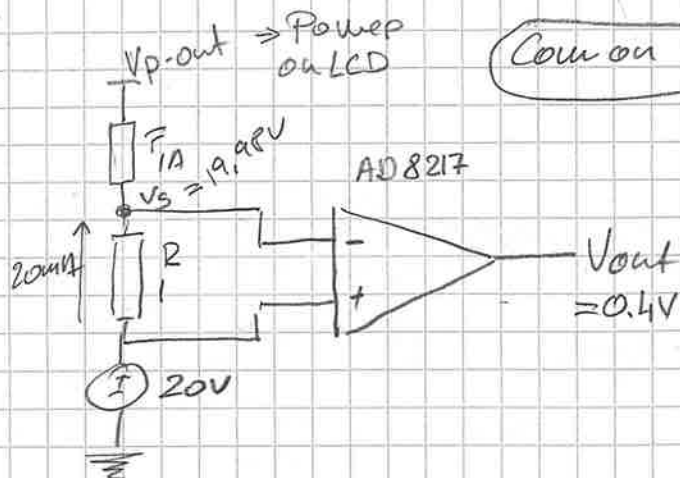


Material $D_k = \epsilon_r \rightarrow \epsilon_{eff}$

FR4 $D_k = 4.2$

$$\lambda_g = \frac{c}{f \sqrt{\epsilon_{eff}}}$$

— dielectric permittivity constant



Common Mode Op

$$V = I \cdot R$$

$$20 \mu A = 0.02 A$$

$$20 - V_s = 0.02 A \cdot 1 R$$

$$V_s = 19.98 V$$

Drop voltage!

$$V_{drop} = V_{in} - V_s = 20 V - 19.98 V = 0.02 V$$

$$Output = V_{in} \cdot V_{drop}$$

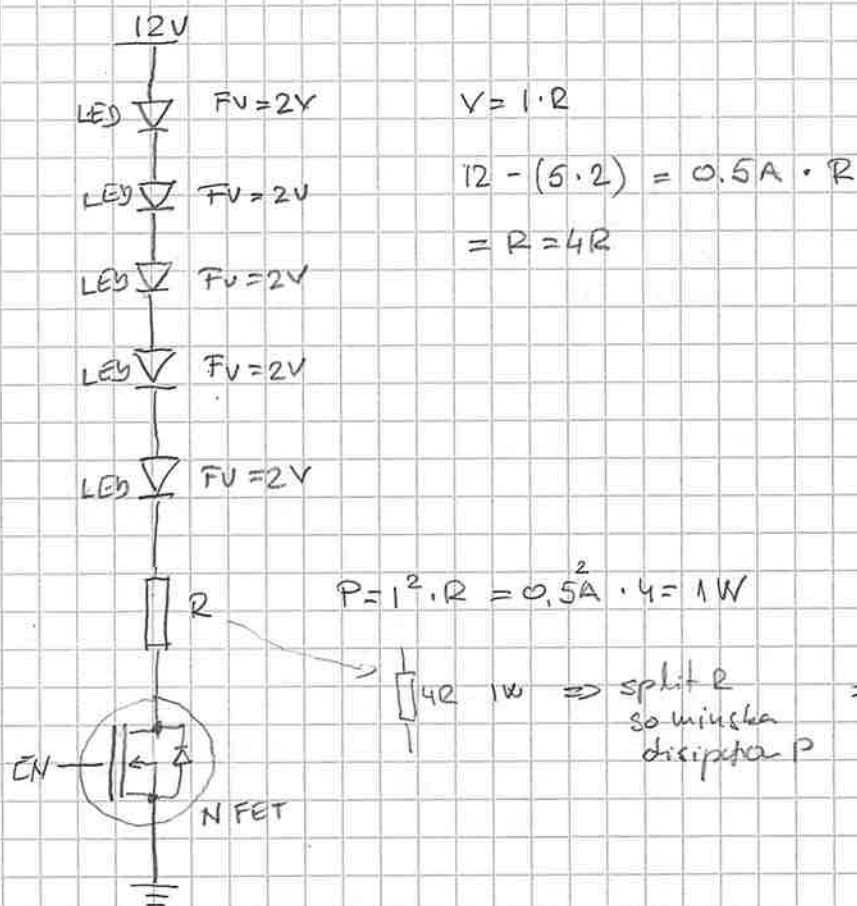
$$= 20 V \cdot 0.02 V = 0.4 V$$

Test circuit

How much current go

To Vp-out

Dioder och Transistor



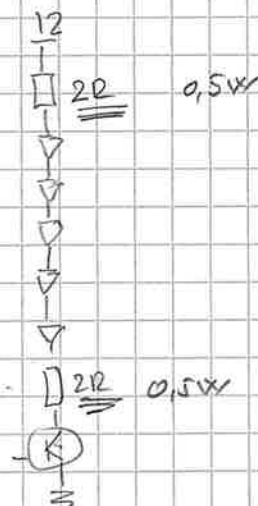
$$V = I \cdot R$$

$$12 - (5 \cdot 2) = 0.5 A \cdot R$$

$$= R = 4 R$$

$$P = I^2 \cdot R = 0.5^2 \cdot 4 = 1 W$$

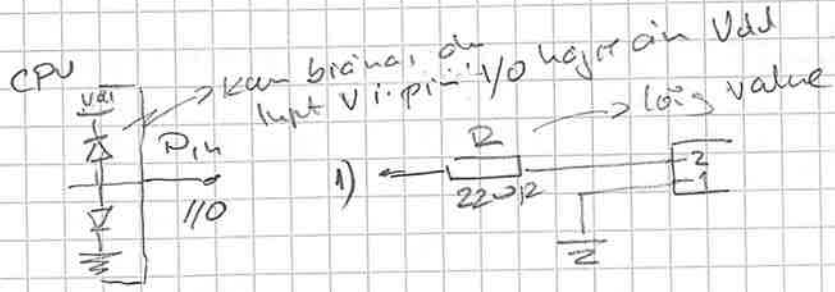
split R
so miniska
dissipation P



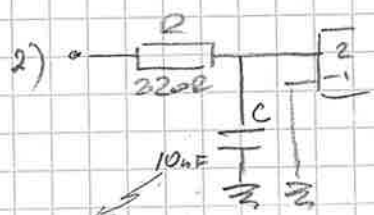
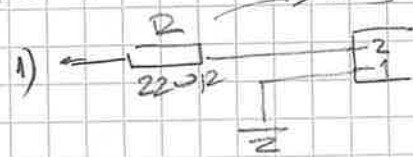
Man kan använda LED DRIVER
då kan man slipa använda
resistorer

switching constant
current (AL3060)

ESD Protection



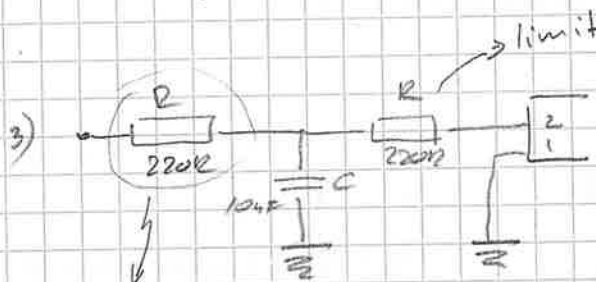
OBS! ESD
ska placera
här kontakt
so nära som
möjligt



Dependent of
frequency speed
höjra pf till 10nF exempel

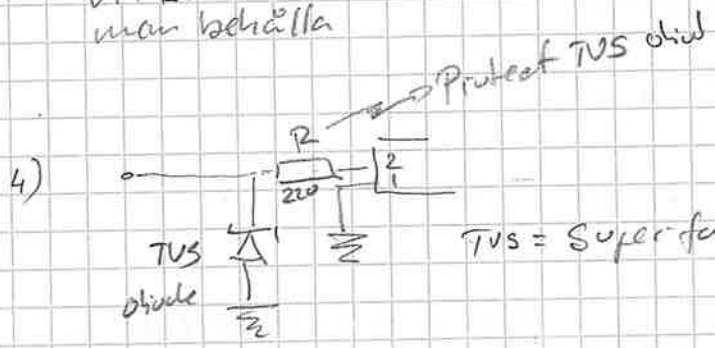


Om det är enkelt
on/off input
Can we connect
1μF



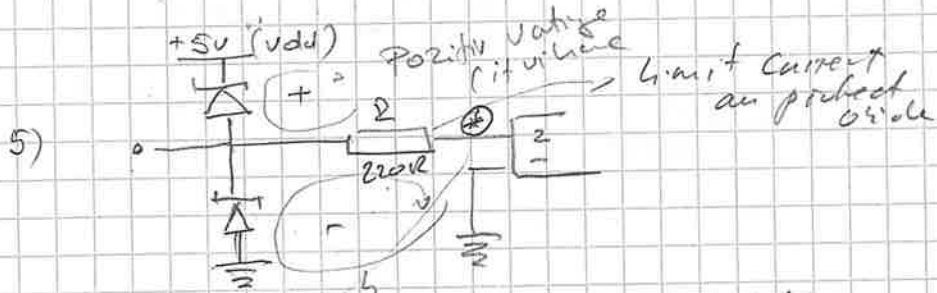
limit current
som gör till C
Protect C

Om man
vill kan
man behålla



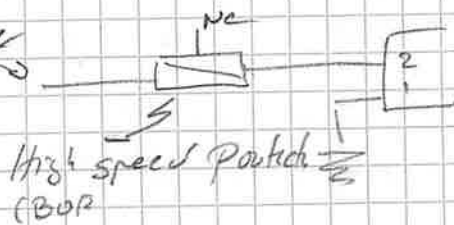
Protect TVS diode

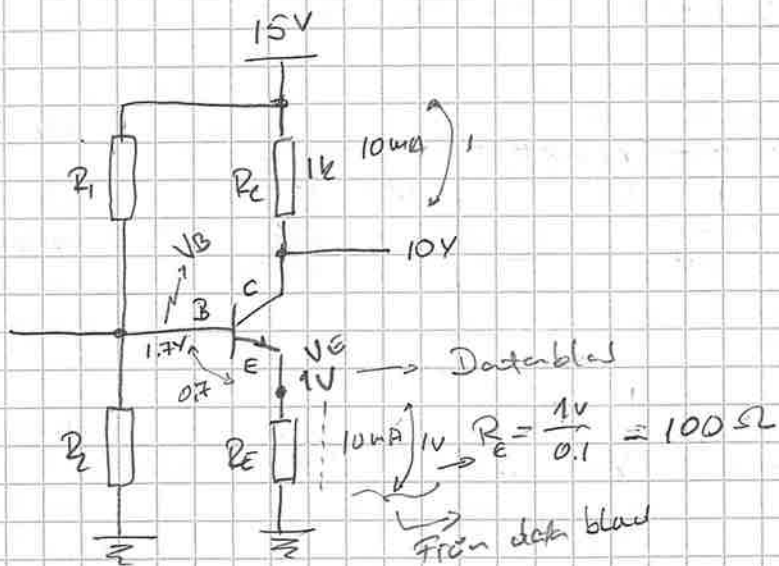
TVS = Super fast diode



if we have negative Voltage

Kan använda
men ibland
behövs inte

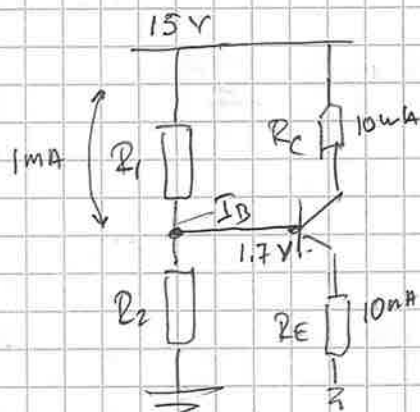




$$V_{BE} = ? \text{ Titta } \Rightarrow V_{BE} = 0.7 \text{ V}$$

$$V_B = V_{BE} + V_E = 0.7 \text{ V} + 1 \text{ V} = 1.7 \text{ V}$$

$$R_1 = ? \text{ och } R_2 = ?$$



$$h_{FE} = \begin{matrix} 100 \\ 250 \\ 300 \end{matrix} \rightarrow \text{Use min value}$$

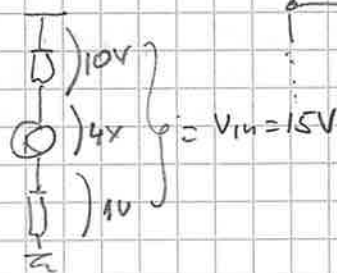
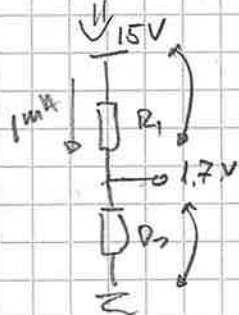
$$h_{FE} = ? \rightarrow \text{Förstärkning}$$

$$1 \text{ mA} : I_B = h_{FE} = I_C$$

$$10 \text{ mA} : R_E \Rightarrow 10 \text{ mA} : R_C$$

$$\Rightarrow \begin{matrix} R_C \\ 1k \end{matrix} \Rightarrow 10 \text{ V}$$

$$I_B = \frac{I_C}{h_{FE}} = \frac{10 \text{ mA}}{10} = 1 \text{ mA}$$

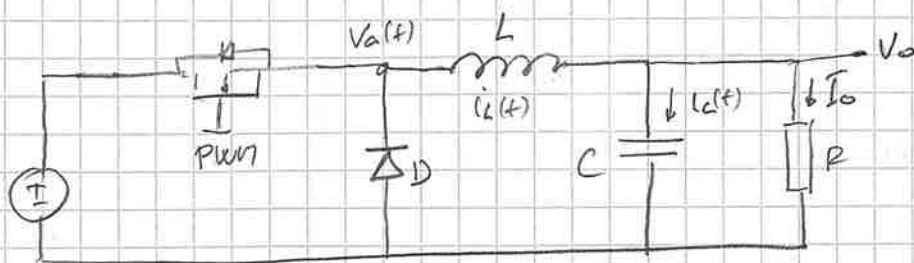


$$\Rightarrow R_1 = \frac{15 - 1.7}{0.001} = 13.3 \text{ k}\Omega = \underline{15 \text{ k}\Omega}$$

$$R_2 = ? \Rightarrow 90\% \text{ g\ddot{o}r to ground} \\ \text{Approximation} = 1 \text{ mA}$$

$$\Rightarrow R_2 = \frac{1.7}{0.001} = 1.7 \text{ k}\Omega$$

Buck Converter Design



Key points:

- Use input and output voltage to estimate the duty cycle
- sized inductor
- sized capacitor
- sized diode and use Schottky diode for low loss
- sized the MOSFET and estimate on/off switch time

Design requirement: $V_{in} = 15V$ (nominal)
 $V_{out} = 5V$ (nominal)
 $P_{out} = 10W$ (nominal)
 $\text{Efficiency} = 90\%$

$$I_o = \frac{10W}{5V} = 2A$$

$$V_o = V_i \cdot D \Rightarrow \frac{V_o}{V_i} = D \quad \text{Duty cycle}$$

Switch mode power supply SWPS
 Conduct conduction mode CCM

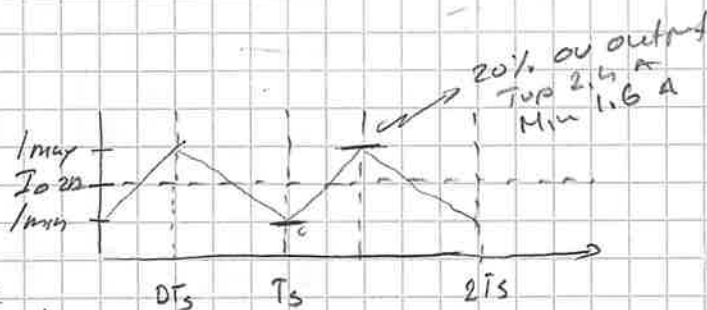
$$\frac{5V}{15V} = \frac{1}{3} = 0.33 = \underline{\underline{33\%}}$$

INDUCTOR SIZE

$$\frac{\Delta I}{I_o} < 40\%$$

$$\Delta I = (0.2)(2A) = 0.4A$$

→ Ripple current



Higher switching frequency
 = Lower inductor
 ⇒ smaller inductor size

$$\Rightarrow \text{upshote } f_s = 500 \text{ kHz}$$

$$L = \frac{V_o(1-D)}{f_s \Delta I} = \frac{5(2/3)}{0.5 \cdot 10^6 \cdot 0.4} = 16.6 \mu H$$

$$\text{Databook: } 22 \mu H, I_{rms} = 6A, 72m\Omega$$

Decrease efficiency
 ⇒ higher ripple current

16.6 μH	20% efficient output	Dec resistance
15 μH	↑ efficient output	50 m Ω → Help to
22 μH		72 m Ω

CAPACITOR SIZE

$$C = \frac{\Delta I_o}{8 f_s \Delta V_o}$$

Estimate $5V \pm 0.1V$

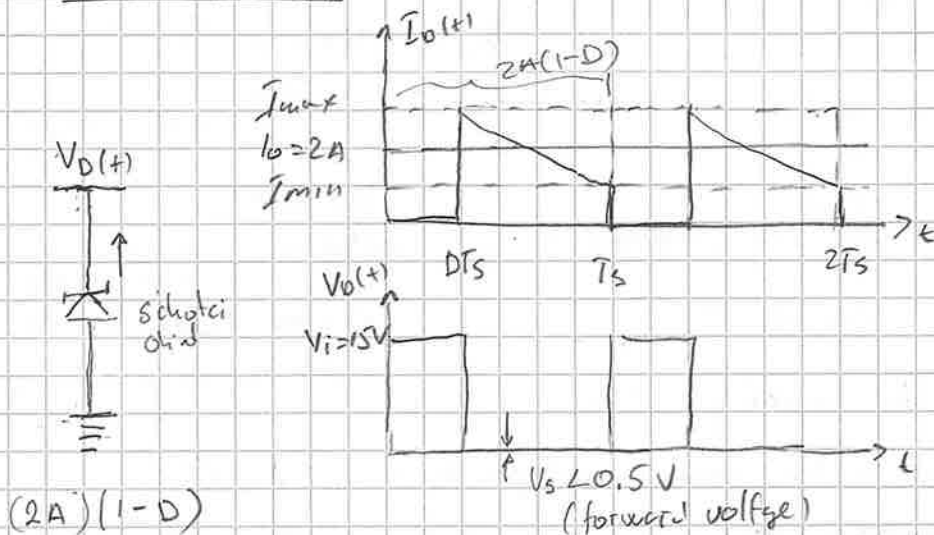
$$\Delta V_o = 0.2V$$

$$C = \frac{0.4}{8(0.5 \cdot 10^6)(0.2)} = 5 \mu F$$

Typical size of capacitor: $4.7 \mu F, 55V_{dc}$

TDR, MLCC

DIODE SIZE

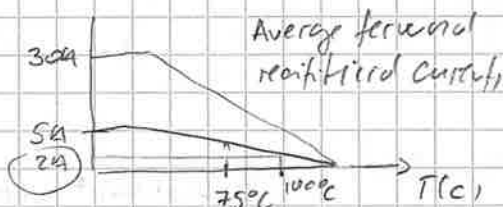
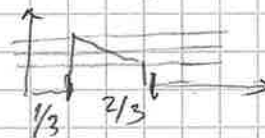


Max reverse voltage is 15V

Not transistors is open: The later 2/3 rds for this example

Diode current $V_f = 0.03V$ (at)

$$I_f = 5A$$



MOSFET SIZE

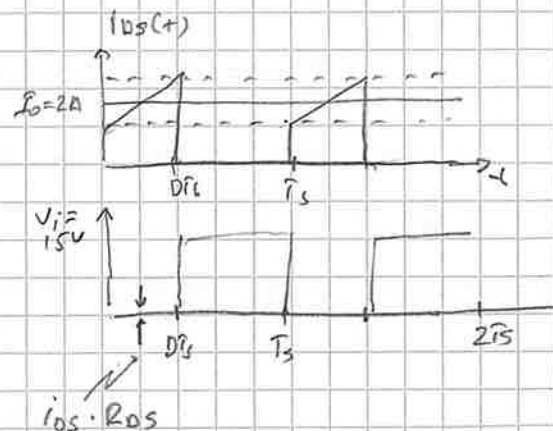


$$I_o \cdot D = 2/3 A$$

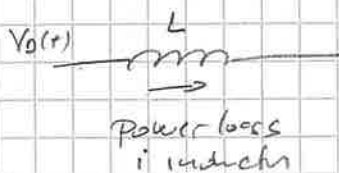
$$V_{DS} > 15V$$

- T_j - below or maximum
- DC/DC converter applies

Rise/Fall time $\rightarrow$ on/off
 Loss then $< 21ns$



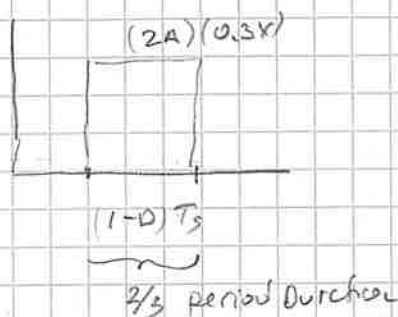
Insertion loss i inductor



$$L = 16.5 \mu H \quad \text{Datasheet } 15 \mu H, R_{ESR} = 50 \Omega$$

$$P_L = I_0^2 \cdot R_{ESR} = (2A)^2 \cdot 50 m\Omega = 200 mW$$

Insertion loss i diode



$$V_f = 0.3 V$$

$$P_D = (2A)(0.3V) \cdot \frac{2}{3} = 400 mW$$

Duration of period

Insertion loss i MOSFET

2 losses for Mosfet:

Key Points

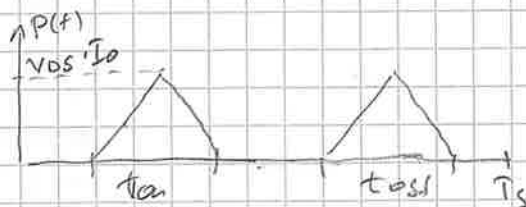
- Developed a power budget 1.1 W
- Estimated loss 200mW, 400mW, 180mW
- Validated the efficiency of the converter 79%

• Conduction loss $\rightarrow$ dependent of $I_0 = 2A$

• Switching Loss $\rightarrow$ dependent of I_{on} , I_{off} , f_s , V_{DS} , I_0

$$P_{cond} = \frac{I_0^2 R_{DS} \cdot D f_s}{f_s} = (2A)^2 \cdot 8.5 m\Omega \cdot \frac{1}{3} = 11.3 mW$$

Switching:



$$P_{sw} = \frac{\frac{1}{2} V_{DS} I_0 (t_{on} + t_{off})}{T_s} = \frac{1}{2} V_{DS} I_0 (t_{on} + t_{off}) f_s$$

$$= \frac{1}{2} (15V)(2A) (21ns) (0.5 \cdot 10^6)$$

$$P_{sw} = 15 (21 \cdot 10^{-9}) (0.5 \cdot 10^6) = 157 \cdot 10^{-3} W = 157 mW$$

$$P_{MOSFET} = (P_{cond} + P_{sw}) = 11.3 + 157 = 168.3 mW$$

$$P_{TOTAL LOSS} = \underbrace{200 mW}_{\text{Inductor}} + \underbrace{400 mW}_{\text{diode}} + \underbrace{168.3 mW}_{\text{Mosfet}} = 768 mW < 1100 mW$$

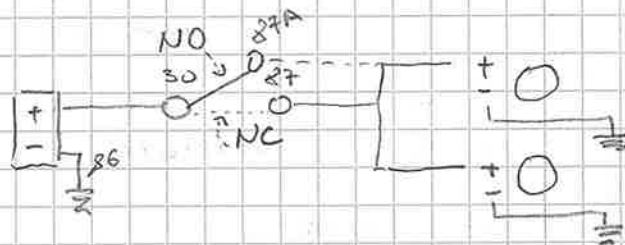
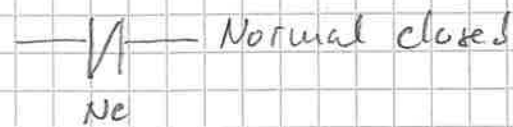
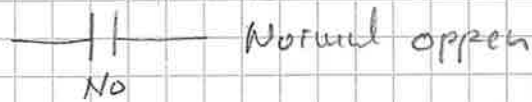
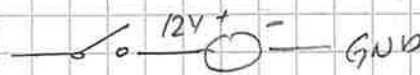
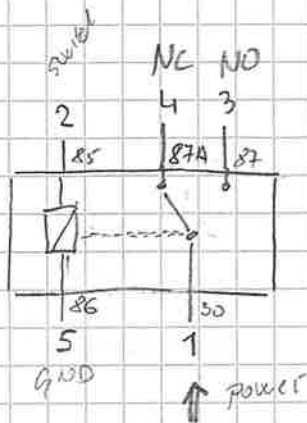
$$\eta_{synch} \approx \frac{10W}{10.54W} = 0.948$$

$$= 94.8\%$$

$10W + 0.54W$

$$P_{synch loss} = 200 mW + 2(168.3 mW) = 536.6 mW$$

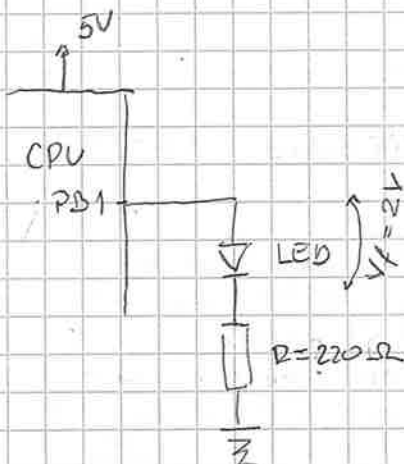
Relay



"ON" → Normal: open is closed ⇒ "ON"
⇒ gate pin 87A

→ Normal: closed is open ⇒ "OFF"
⇒ gate pin 87A

LED

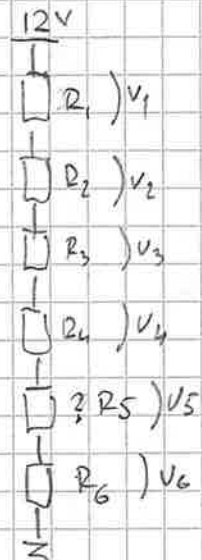


$$V = IR$$

$$5 - 2 = I \cdot 220 \Omega$$

$$\Rightarrow I = 13 \text{ mA}$$

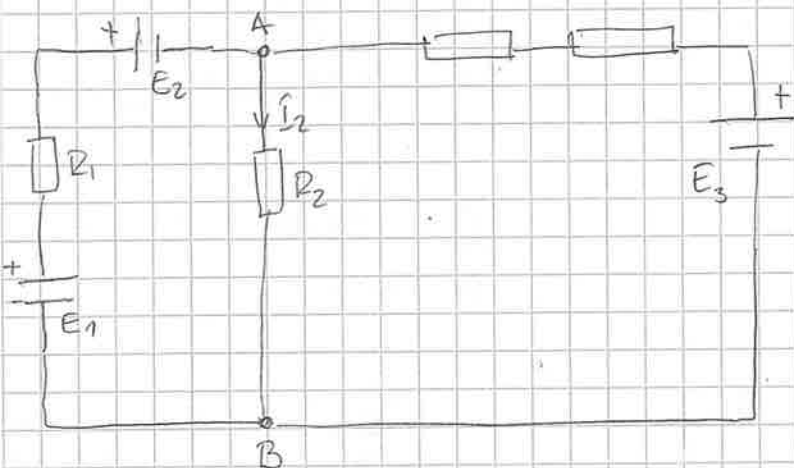
EX



$$\Rightarrow V = I \cdot R$$

$$12V - V_1 - V_2 - V_3 - V_4 - V_6 = \frac{V_5}{I \cdot R_6}$$

beräkning av teorin



$$E_1 = 18V$$

$$E_2 = 12V$$

$$E_3 = 24V$$

$$R_1 = 1\Omega$$

$$R_2 = 1\Omega$$

$$R_3 = 3\Omega$$

$$R_4 = 7\Omega$$

$$I_2 = ?$$

STEG

R_2 (mellan A och B) tas bort

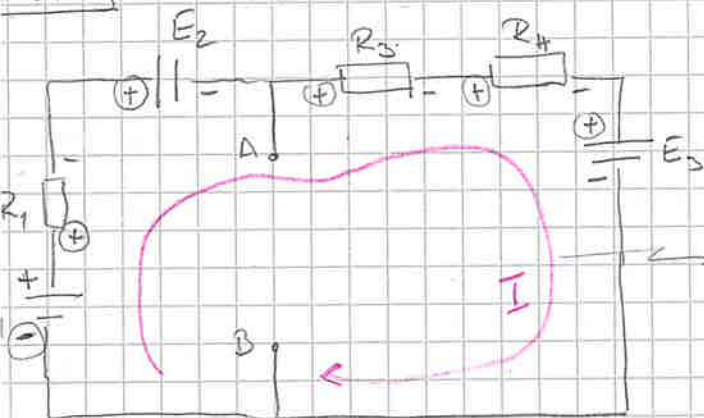


E_2 och E_3 kortslutade

$$R_T = R_1 // (R_3 + R_4)$$

$$R_T = \frac{R_1(R_3 + R_4)}{R_1 + R_3 + R_4} = \frac{1 \cdot (3 + 7)}{1 + 3 + 7} = \frac{10}{11} \Omega$$

steg



$$U_T = ?$$

$$U_T = U_{AB} = U_A - U_B$$

vi vet inte vilken riktning för ström men sått en riktning

$\left\{ \begin{array}{l} \text{- om korrekt blir } I = + \\ \text{- om inte korrekt } I = - \end{array} \right\}$
 ström riktning

$$-E_1 + IR_2 + E_2 + IR_3 + IR_4 + E_3 = 0$$

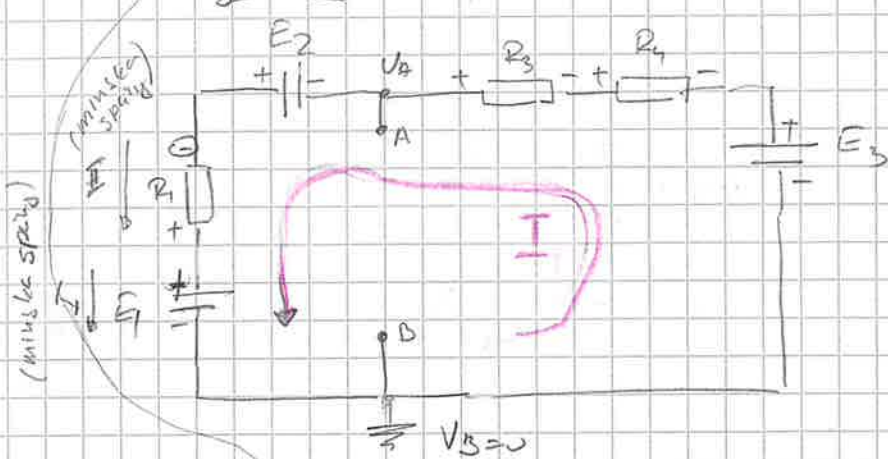
$$\Rightarrow 18 + I(1 + 3 + 7) = 0$$

$$I = -\frac{18}{11}$$

$$I = -1.636$$

Vi ändrar ström riktning så att den blir korrekt:

$I \leftarrow$
 E_2 : na- $\ominus$ till $\oplus$ (oba spring)



$$U_A + E_2 - IR_1 + E_1 = 0$$

$\frac{18}{11} \cdot 1$

$$U_A = 7,636 \text{ V} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} = U_{AB} = U_T = 7,636 \text{ V}$$

$$U_B = 0$$

oba spring

$$\left. \begin{array}{c} E \\ + \quad | \quad - \\ \hline I \end{array} \right\} \Rightarrow +E$$

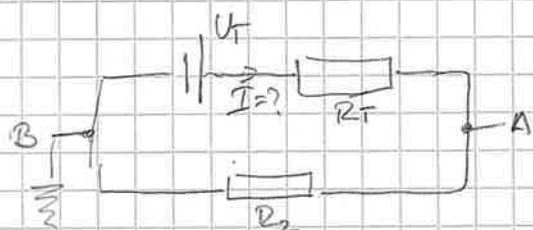
minsko spring

$$\left. \begin{array}{c} E \\ + \quad | \quad - \\ \hline I \end{array} \right\} \Rightarrow -E$$

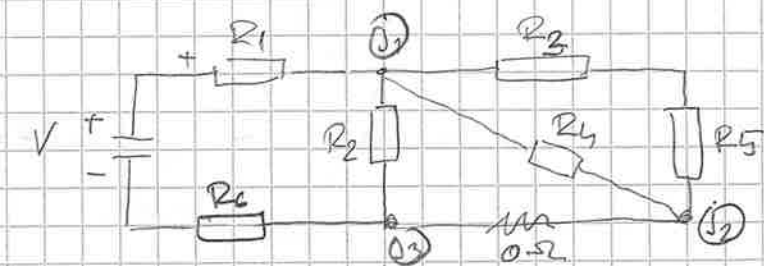
$$\left. \begin{array}{c} R \\ + \quad \text{---} \quad - \\ \hline I \end{array} \right\} \Rightarrow IR$$

$$\left. \begin{array}{c} R \\ + \quad \text{---} \quad - \\ \hline I \end{array} \right\} \Rightarrow -IR$$

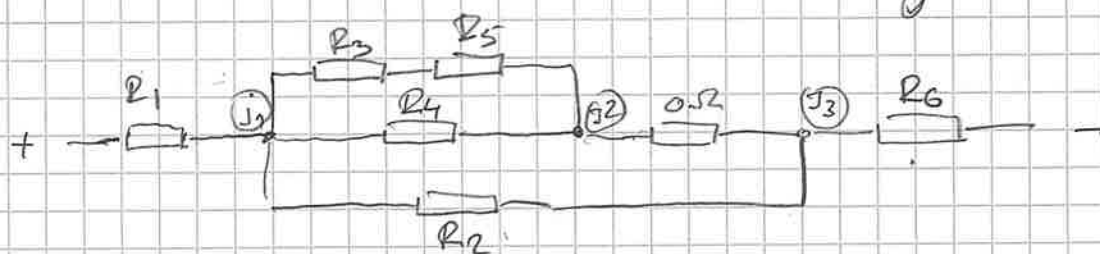
3 steps



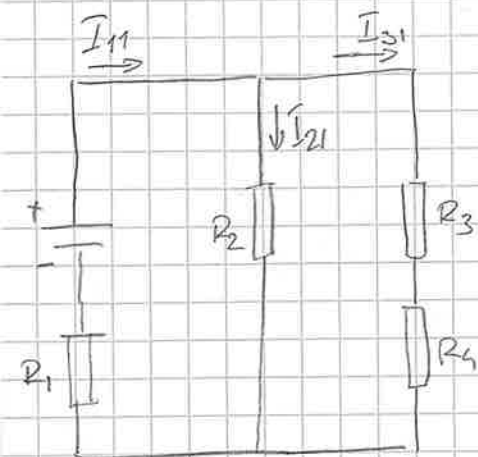
$$I = 2 = \frac{U_T}{R_T + R_2} = 4 \text{ A}$$



shuv ou



superposition

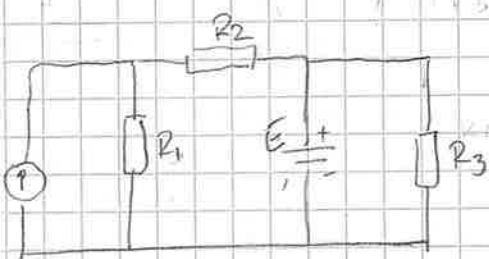


$$I_{21} = \frac{R_{34}}{R_2 + R_{34}} I_{11} \quad \vee$$

$$I_{31} = \frac{R_2}{R_2 + R_{34}} I_{11}$$

$$I_{11} = \frac{E_1}{R_{\text{ukl}}} = \frac{E_1}{R_1 + R_2 // (R_3 + R_4)}$$

superposition



$R_1 = 70 \Omega$
 $R_2 = 70 \Omega$
 $R_3 = 140 \Omega$
 $E = 35 \text{ V}$
 $I = 2 \text{ A}$

$I_5 = 0$ (same potential
 $\rightarrow$ no current)

$I_1' = 2$

$I_3' = 1$

$I_2' = 1$

$I_4' = 1$ (same as I_3)

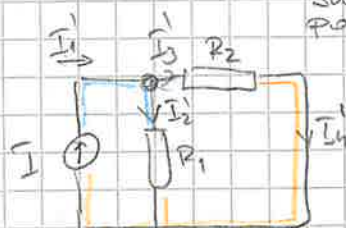
$E = \text{uncoupled}$

$E \rightarrow \text{arbit} \Rightarrow$



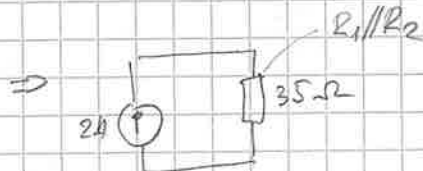
same potential

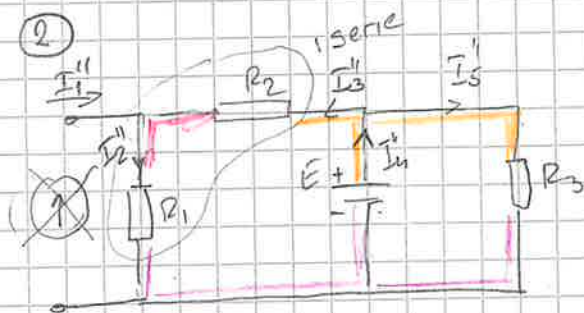
$\Rightarrow$



$I_1' = I_3' + I_2' = 2$

$R_1 // R_2$

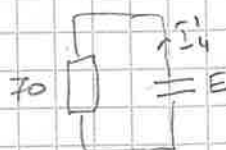




ström källa I urkopplat

$$I_1'' = 0$$

$$R_{\text{total}} = (R_1 + R_2) \parallel R_3 = \frac{140 \cdot 140}{140 + 140} = 70$$

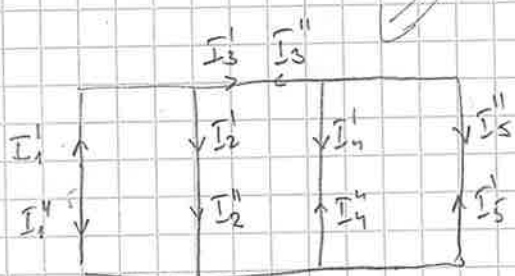


$$I_4'' = \frac{35}{70} = 0,5 A$$

$$I_3'' = I_2'' = \frac{E}{R_1 + R_2} = 0,25 A$$

$$I_5'' = \frac{E}{R_3} = \frac{35}{140} = 0,25 A$$

③ Superposition // tita skellet och add strömmar från ① och ②



$$I_5'' = 0,25 \quad I_5' = 0$$

$$I_1'' = 0 \quad I_1' = 2$$

$$I_2'' = 0,25 \quad I_2' = 1$$

$$I_3'' = 0,25 \quad I_3' = 1$$

$$I_4'' = 0,5 \quad I_4' = 1$$

$$I_1 = 1 - 0,5 = 0,5 A$$

$$I_3 = 1 - 0,25 = 0,75 A$$

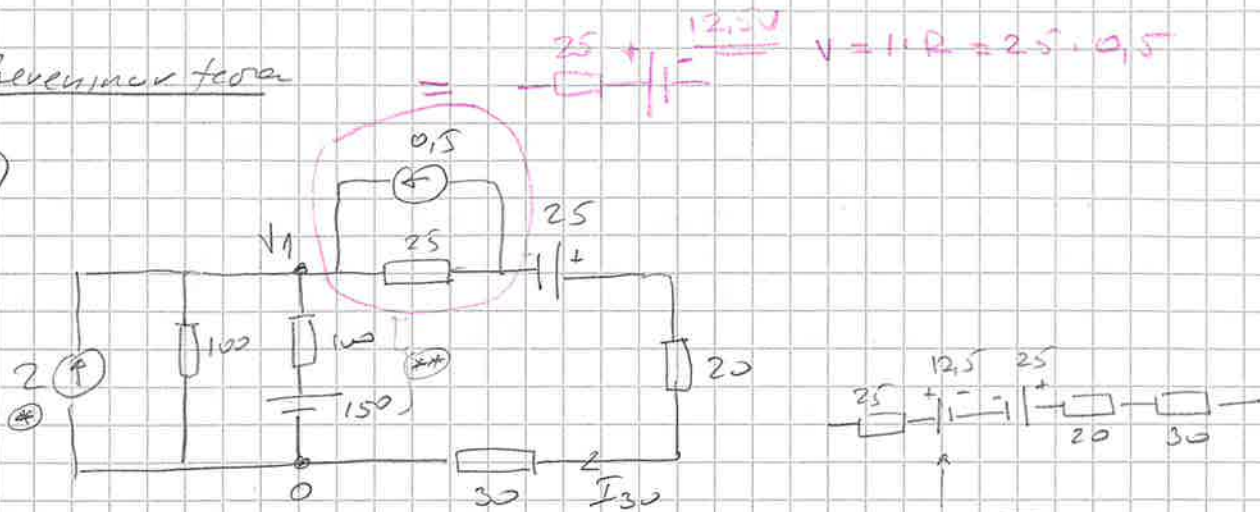
$$I_2 = 1 + 0,25 = 1,25 A$$

$$I_1 = 2 A$$

$$I_5 = 0,25 A$$

Thevenin's theorem

1)



$$V_1 \left(\frac{1}{100} + \frac{1}{100} + \frac{1}{25+30+25} \right) = \frac{2}{100} + \frac{1,5}{100} + \frac{25-12,5}{25+20+30}$$

$\frac{V}{R} = \frac{150}{100} = 1,5$

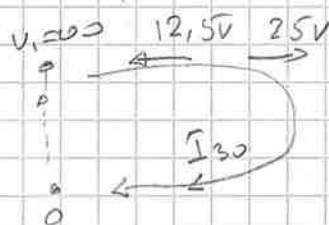
$$V_1 \cdot \frac{3,2}{150} = \frac{20}{6}$$

$$V_1 = 100V$$

$$I_{30} = \frac{V_1 + 12,5}{25+20+30} = \frac{112,5}{75}$$

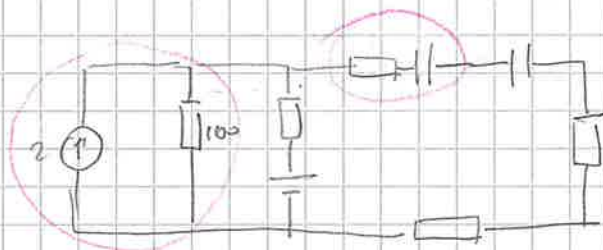
$$I_{30} = 1,5A$$

Stromfluss

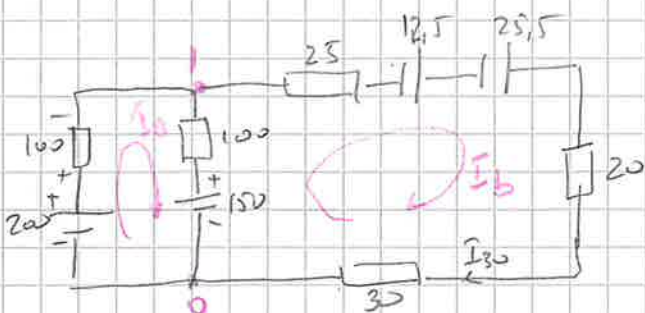


$$100 - 12,5 + 25 = 112,5V$$

2)



$$V = I \cdot R = 200$$



$$I_{30} = I_D$$

$$200I_A - 100I_B = 200 - 150$$

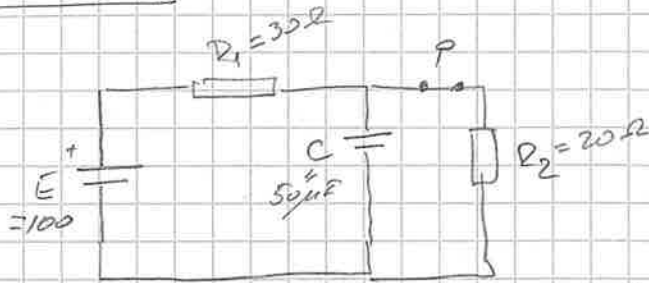
$$V = I \cdot R$$

$$175I_B - 100I_A = 150 - 12,5 + 25$$

$$100+25+20+30$$

$$V = I \cdot R$$

RC SP20

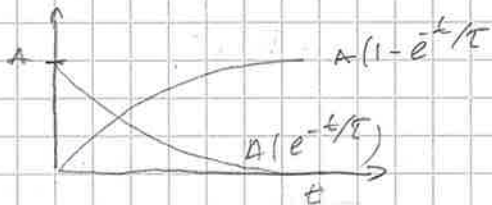


Thévenin: $R_T = R_1 // R_2 \Rightarrow$



$$E_T = \frac{R_2}{R_1 + R_2} E \Rightarrow E_T = 40V$$

$$E_T = 40V$$



$$V_C = 60(1 - e^{-t/\tau}) + 40V$$

amplitude (100V - 40V = 60V)

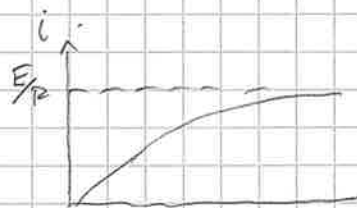
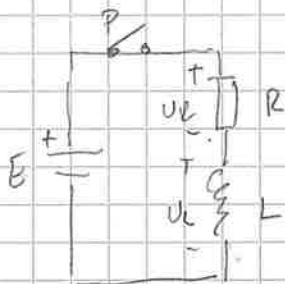
$$\tau = R_T \cdot C = 30 \cdot 50 \cdot 10^{-6} = 1.5 \text{ms} = \frac{3}{2} \cdot 10^{-3}$$

$$V_{R_1} + V_C = E$$

$$V_{R_1} = E - V_C$$

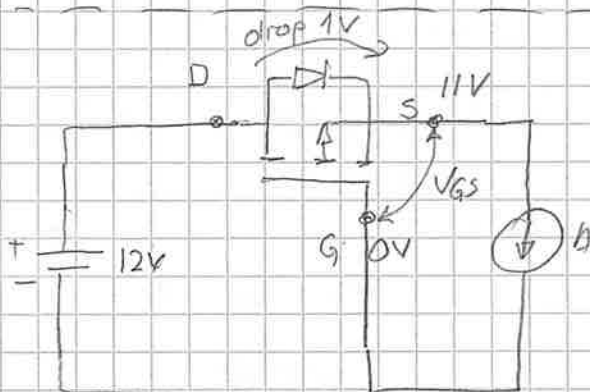
$$V_{R_1} = 60e^{-666.6t}$$

RL SP20



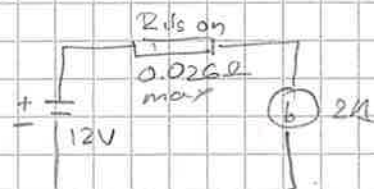
$$i = \frac{E}{R}(1 - e^{-t/\tau})$$

$$\tau = \frac{L}{R}$$



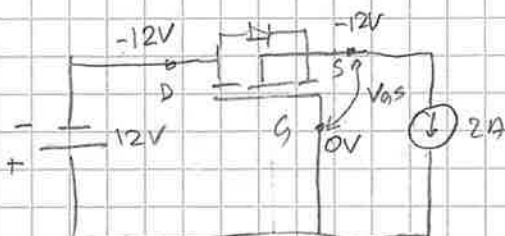
TRANSISTOR MOSFET

$$V_{GS} = V_G - V_S = 0V - 11V = 11V$$



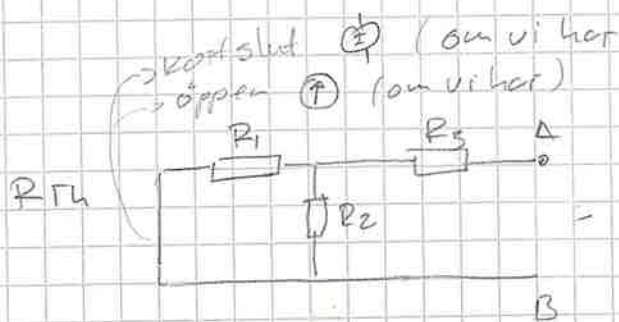
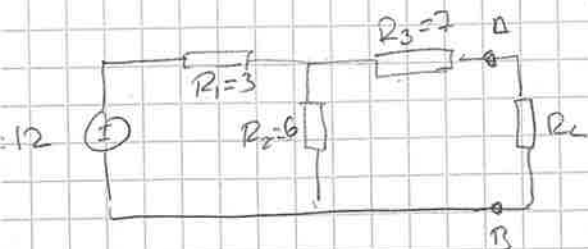
Power loss:

$$P_R = I^2 \cdot R = 2A \cdot 2\Omega \cdot 0.026\Omega = 0.104W$$



$$V_{GS} = V_G - V_S = 0V - (-12V) = 12V$$

von Transformativ

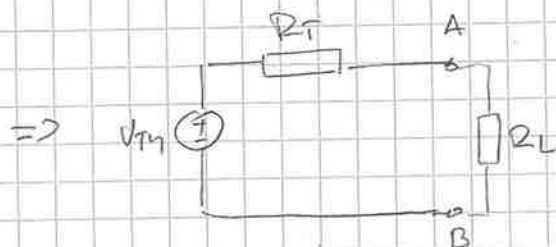


$$R_{Th} = R_1 // R_2 + R_3$$

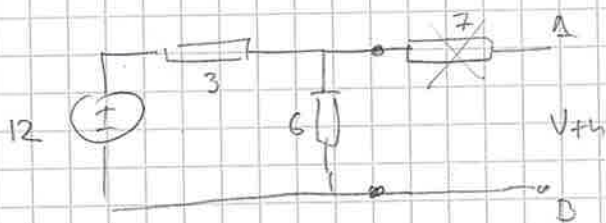
$$R_{Th} = \frac{R_1 R_2}{R_1 + R_2} + R_3$$

$$= \frac{3 \cdot 6}{3 + 6} + 7$$

$$= 2 + 7 = \underline{\underline{9}}$$

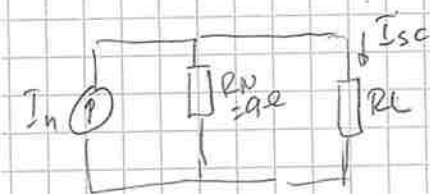


$$V_{Th} = V_{AB} =$$

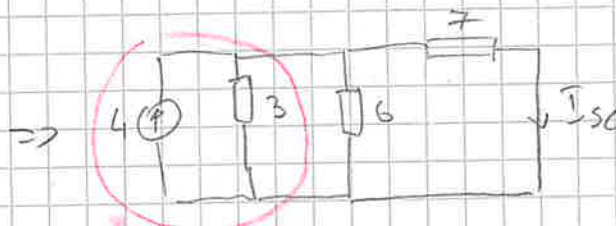
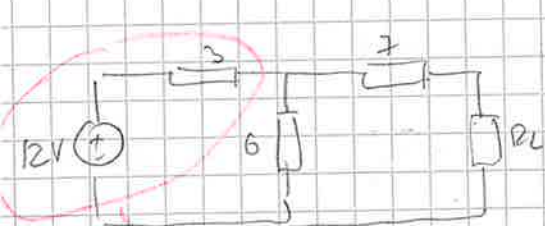


$$\Rightarrow V_{Th} = \frac{6}{3+6} \cdot 12 = 8V$$

Northern equivalent:

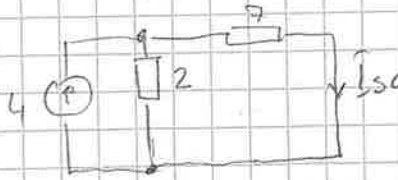
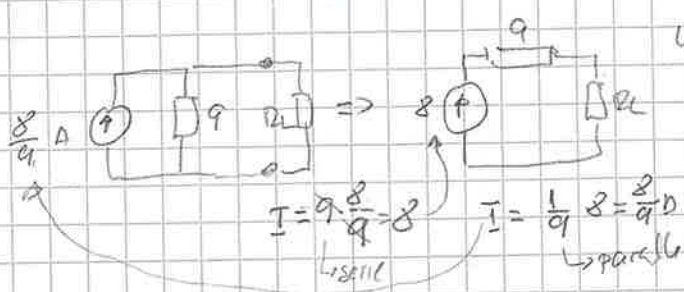


$$I_N = I_{sc}$$



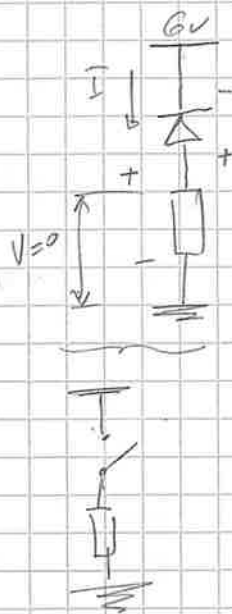
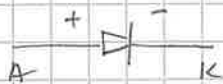
$$R_{Th} = R_N = 2$$

$$I = \frac{V}{R} = \frac{12}{3} = 4$$



$$I_{sc} = I_N = 4 \left[\frac{2}{2+7} \right] = 4 \cdot \left[\frac{2}{9} \right] = \frac{8}{9} A$$

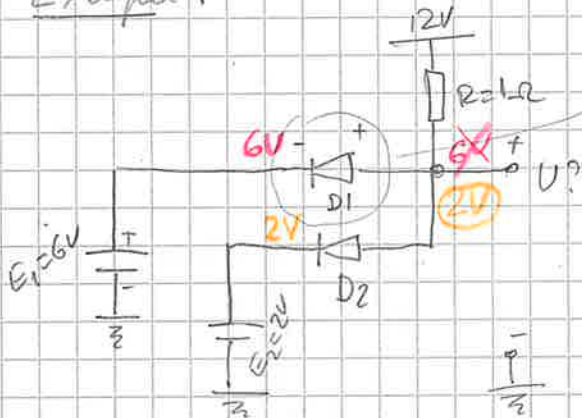
Diode



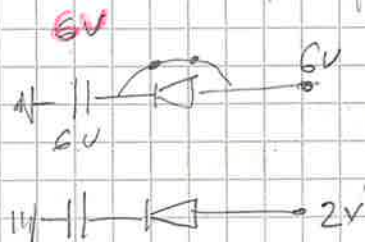
$I = \frac{6V}{6k\Omega} = 1mA$
(Ideal Diode in series with a voltage source)



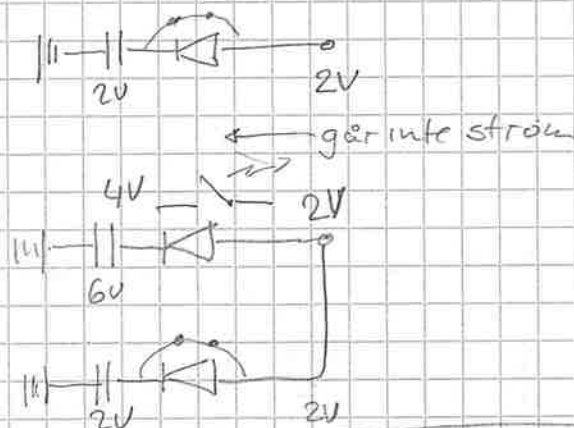
Exempel:



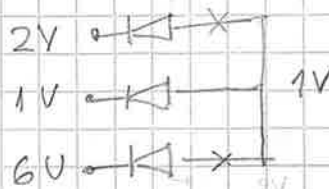
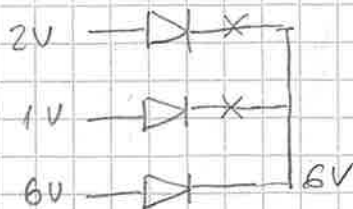
1. D1 directly polarized?



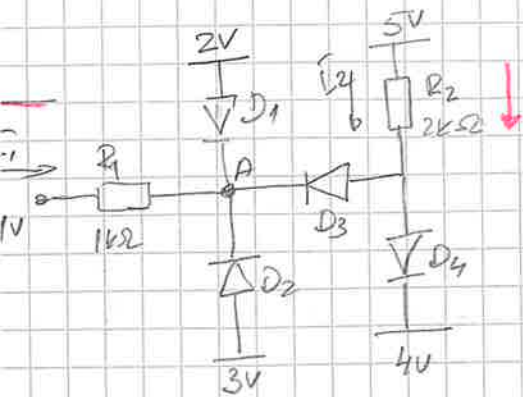
2. D2 directly polarized?



Invert polarized $\Rightarrow$ 1mA ström



Bara on diod
kan

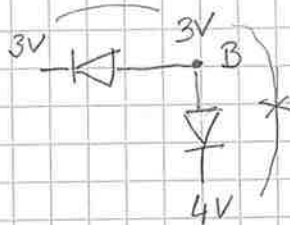
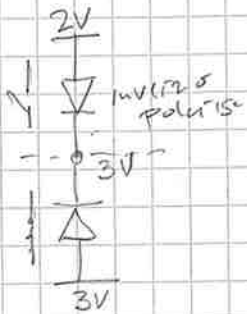


$$V_A = 3V$$

$$V_B = 3V$$

$$I_1 = \frac{1-3V}{1k\Omega} = -2\mu A$$

$$I_2 = \frac{5-3V}{2k\Omega} = 1\mu A$$



Anode = Different V
Kathode = sama V

Bara oleh
diod sa har
sterst V

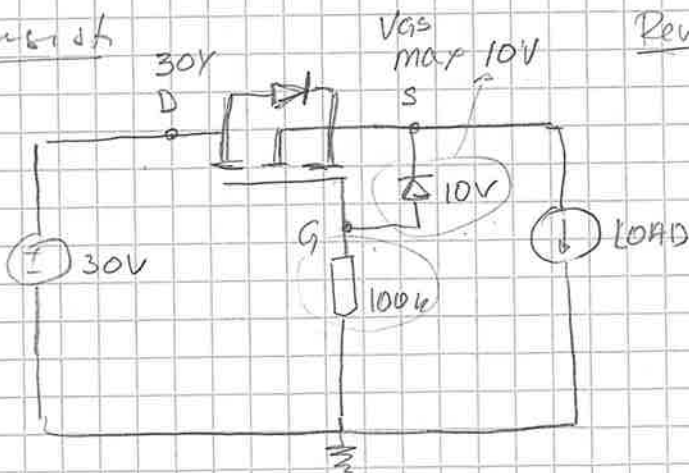


Anode = sama V
Kathode = Different V



Bara oleh sa
har minst V

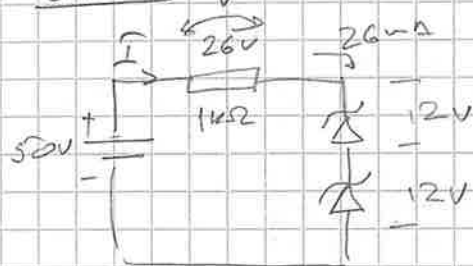
ambish



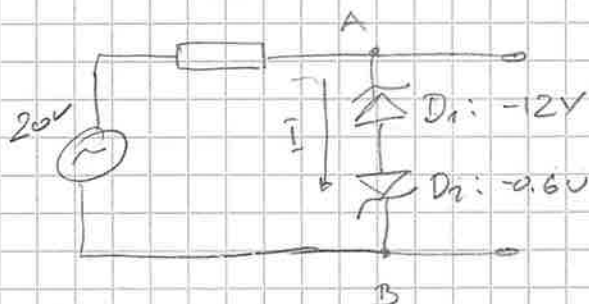
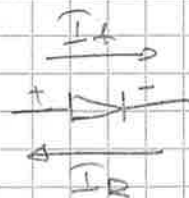
Reverse voltage protection

$$V_{GS} = \pm 15V \text{ max}$$

Diode

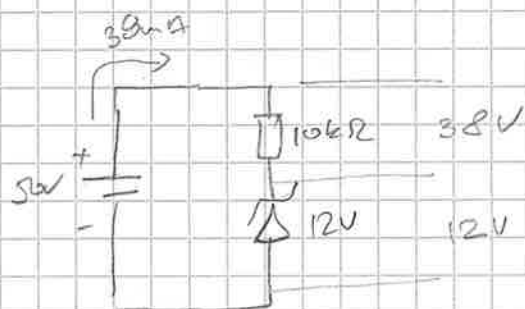
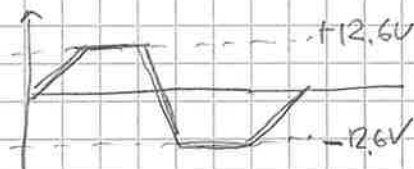
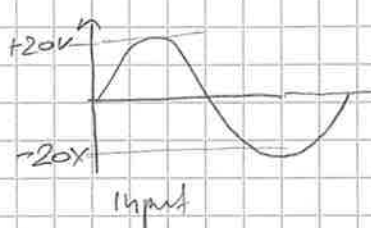


$$I = \frac{50 - 24}{1k\Omega} = 0.026A$$

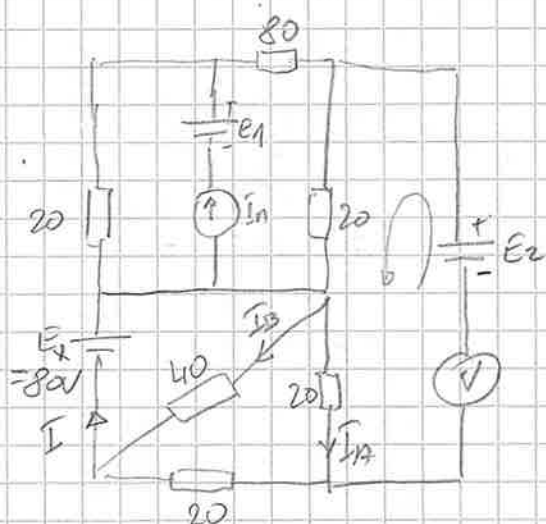


opozit current direction

current direction



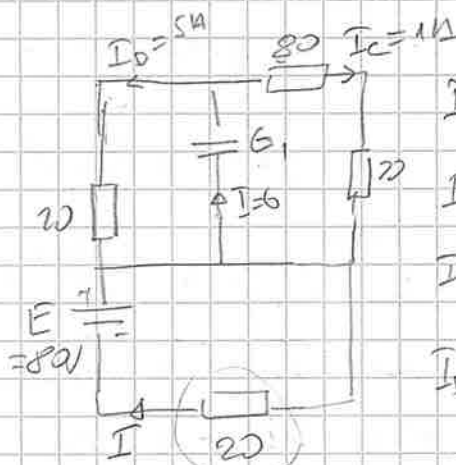
$$I = \frac{50 - 12}{1k\Omega} = \frac{38}{1k\Omega} = 38mA$$



$$I_A = \frac{40}{40+20+20} \cdot I$$

$$= \frac{40}{80} \cdot 4 = 2A$$

$$I_B = \frac{20+20}{40+20+20} \cdot 4 = 2A$$



$$I_C = \frac{20}{20+80+20} \cdot 6A$$

$$I_C = \frac{20}{120} \cdot 6 = 1A$$

$$I_D = \frac{80+20}{(80+20)+20} \cdot 6A$$

$$I_D = \frac{100}{120} \cdot 6 = 5A$$

$$R = 40 // (20+20) = 20$$

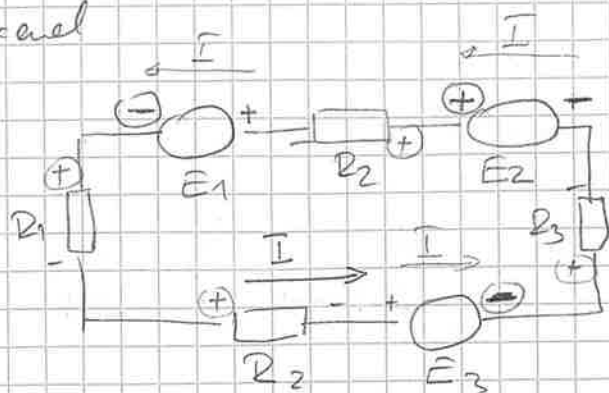
$$\frac{40 \cdot 40}{40+20+20} = \frac{1600}{80} = 20$$

$$I = \frac{80}{20} = 4A$$

$$E_2 = ? \quad \frac{10 \cdot 20}{20} + \frac{I_A \cdot 20}{40} - E_2 = 0$$

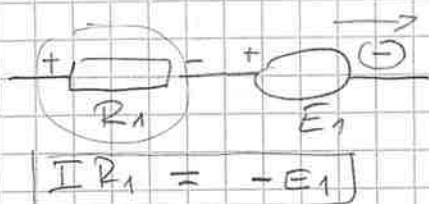
$$\Rightarrow E_2 = 60V$$

anal

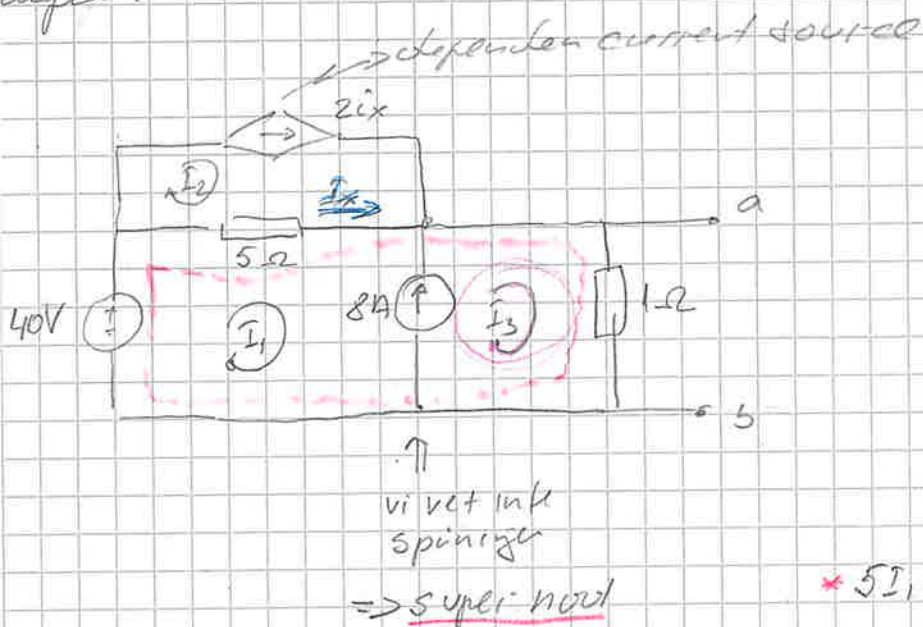


$$I(R_1 + R_2 + R_3 + R_4) = -E_3 + E_2 - E_1$$

$$I = \frac{-E_3 + E_2 - E_1}{R_1 + R_2 + R_3}$$



angul:



① super nod

$$-40 + (I_1 - I_2) \cdot 5 + I_3 \cdot 1 = 0$$

$$-40 + 5I_1 + 5I_2 + I_3 = 0$$

$$* 5I_1 + 5I_2 + I_3 = 40$$

$$* I_3 - I_2 = 8$$

② $I_2 = 2I_x$ AND $I_x = I_1 - I_2$

$$I_2 = 2(I_1 - I_2)$$

$$I_2 = 2I_1 - 2I_2 \Rightarrow * 2I_1 - 3I_2 = 0$$

$$* 5I_1 + 5I_2 + I_3 = 40$$

$$* -I_2 + I_3 = 8$$

$$* 2I_1 - 3I_2 = 0$$

$$AX = B$$

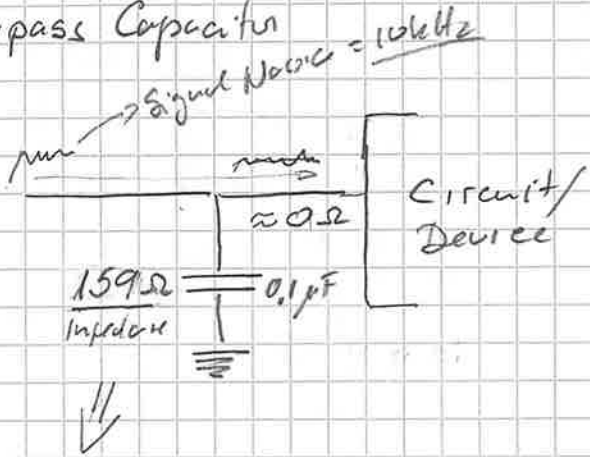
$$\begin{bmatrix} 5 & 5 & 1 \\ 0 & -1 & 1 \\ 2 & -3 & 0 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 40 \\ 8 \\ 0 \end{bmatrix}$$

$$I = A^{-1}B$$

$$\begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 12 \\ 8 \\ 20 \end{bmatrix}$$

$$V_{ab} = \frac{I_3}{1\Omega} = \frac{20}{1} = 20V$$

Bypass Capacitor



High impedance $\Rightarrow$ noise goes to Circuit/Device

$$\text{Capacitor } 0.1 \mu F$$

$$\text{Frequency } 10 \text{ kHz}$$

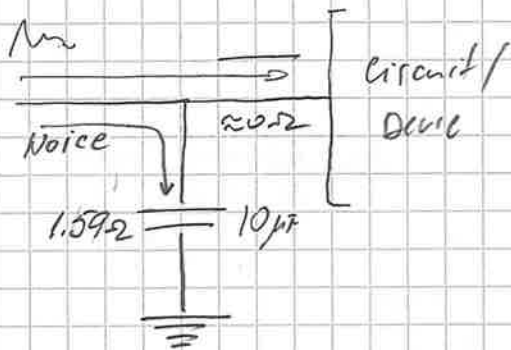
$$\Rightarrow \text{Capacitor impedance } 159.159 \Omega$$

$$\text{Capacitor } 10 \mu F$$

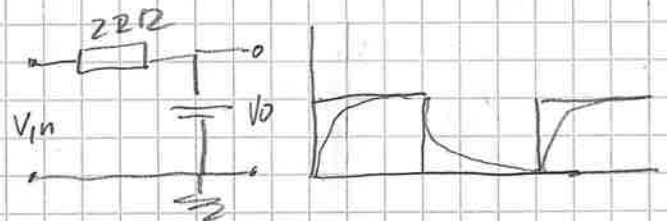
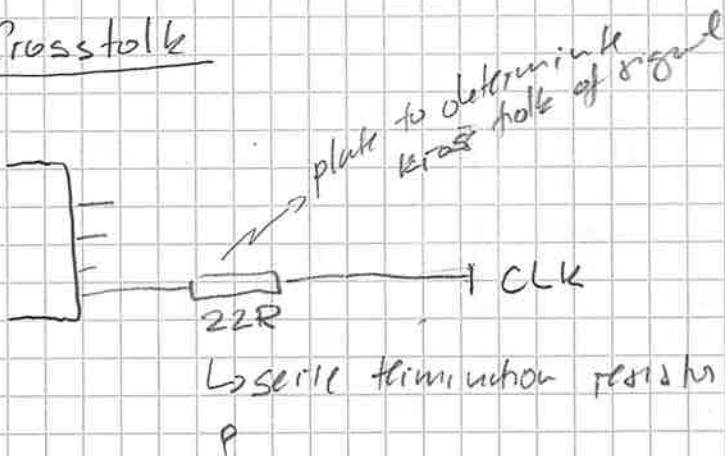
$$\text{Frequency } 10 \text{ kHz}$$

$$\Rightarrow \text{Capacitor impedance } 1.59 \Omega$$

$\Rightarrow$ Bypass capacitor impedance as small as possible

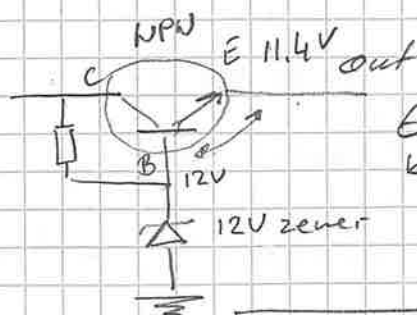


Crosstalk



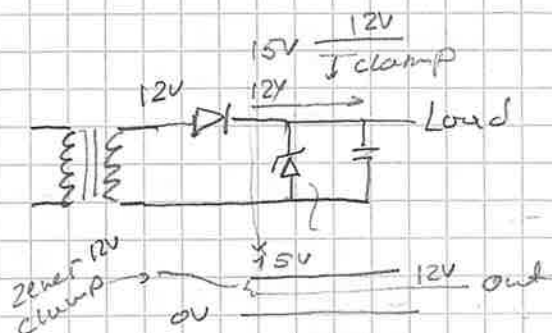
Diode

Transistor is
Emitter follower

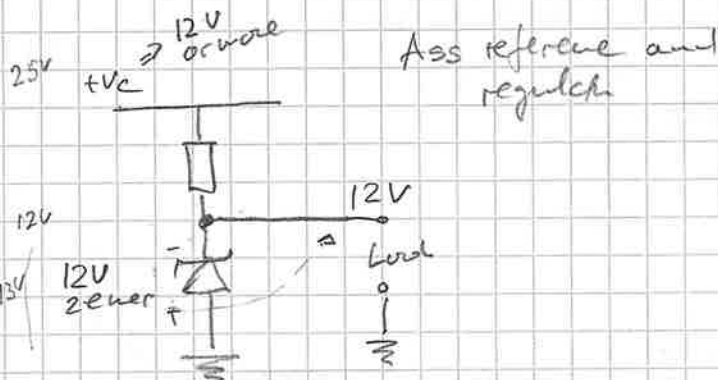
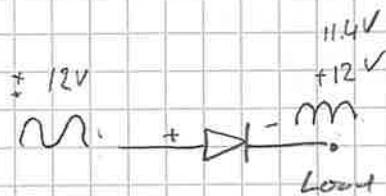
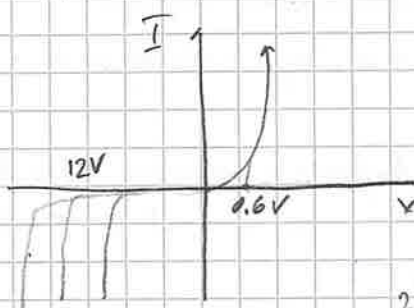


Emitter follow
base

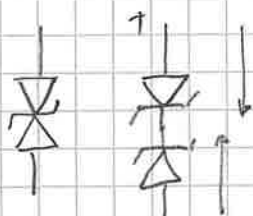
$$V_E = V_B - V_{BE}$$



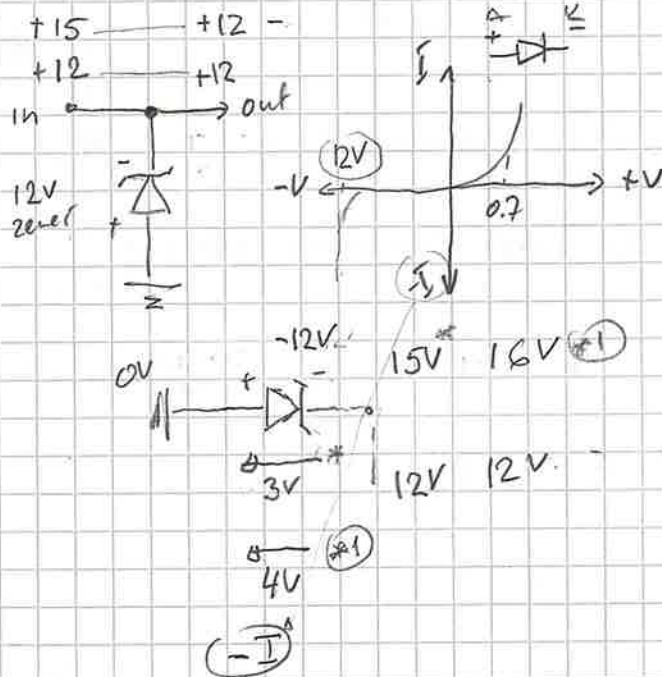
Zener

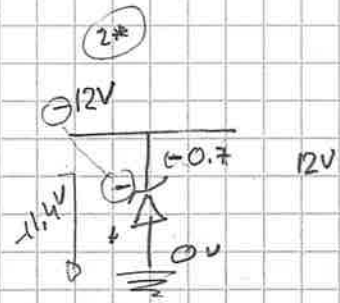
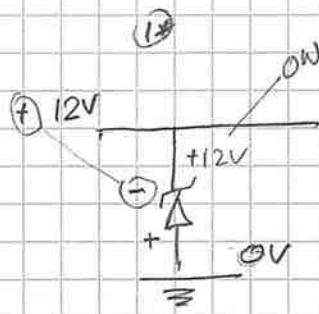
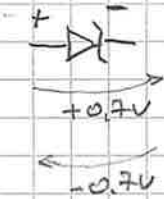
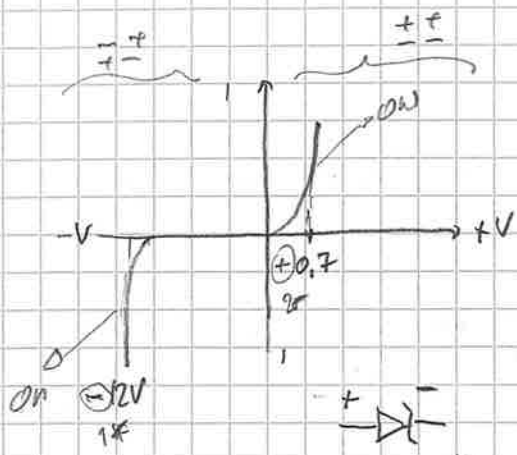


VS



protection of
circuit -
wave vid
TVS diode
Transient suppression
voltage





$$0V \quad 12V$$

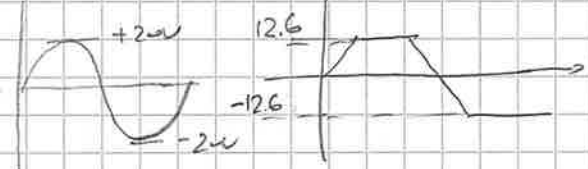
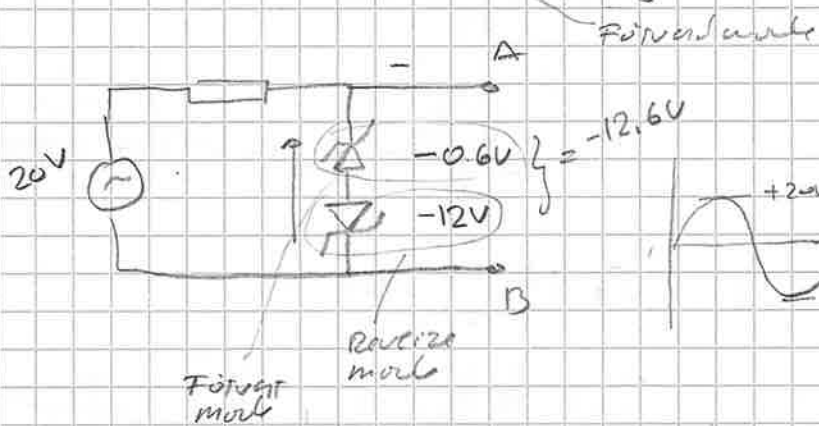
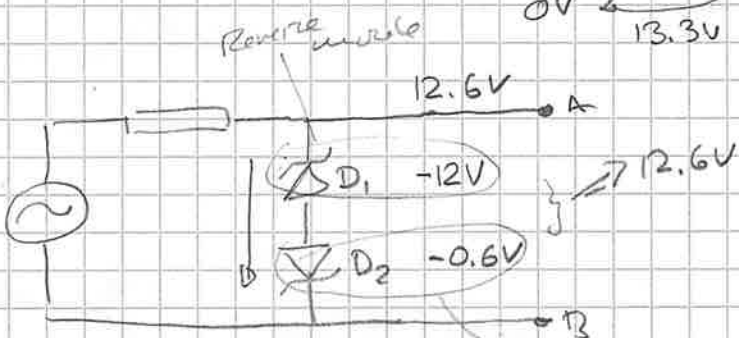
$$0V \quad 3V \quad 15V = 12V$$

$$0V \quad -12V = -0.7V$$

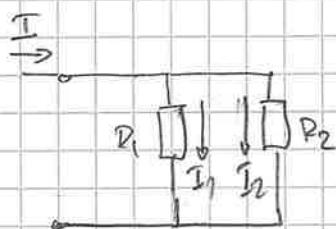
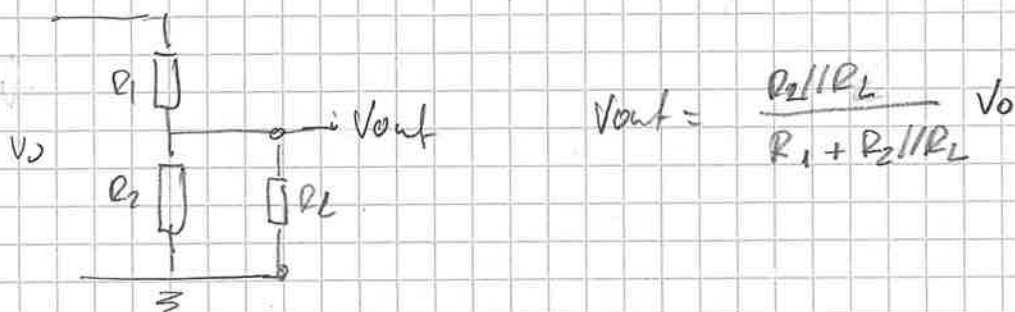
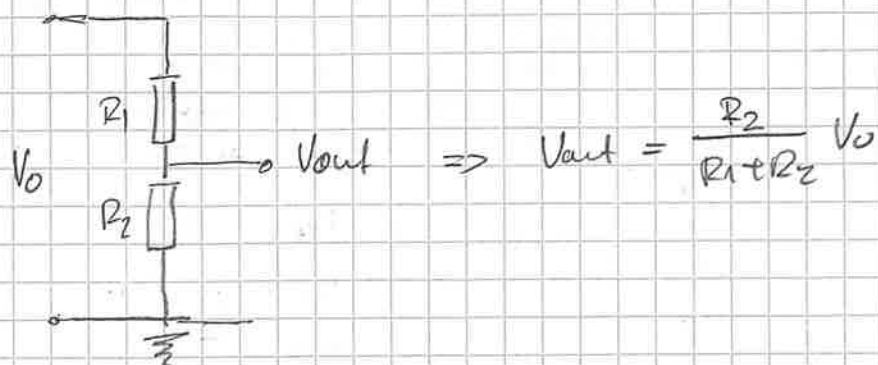
$$-11.3V$$

$$0V \quad -14V = -0.7V$$

$$13.3V$$



law



$$I_1 = \frac{R_{tot}}{R_1} I_{tot}$$

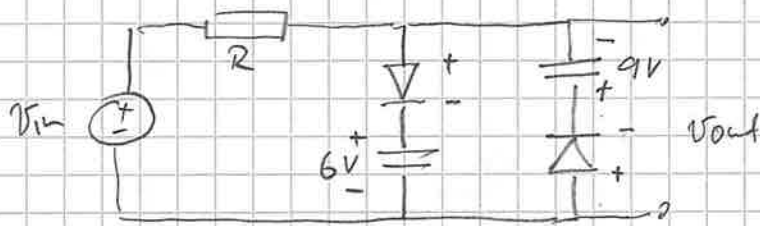
$$R_{tot} = R_1 // R_2 = \frac{R_1 R_2}{R_1 + R_2}$$

$$I_1 = \frac{\frac{R_1 R_2}{R_1 + R_2}}{R_1} I_{tot} \Rightarrow I_1 = \frac{R_2}{R_1 + R_2} I_{tot}$$

$$I_2 = \frac{R_{tot}}{R_2} I_{tot}$$

$$I_2 = \frac{\frac{R_1 R_2}{R_1 + R_2}}{R_2} I_{tot} \Rightarrow I_2 = \frac{R_1}{R_1 + R_2} I_{tot}$$

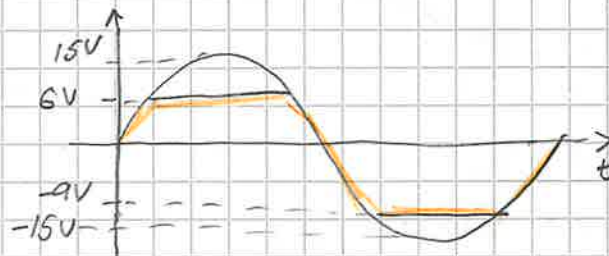
$$v_{in}(t) = 15 \sin(2\pi ft); f = 1 \text{ kHz}$$



Spänningen över en diod kan inte vara positiv!

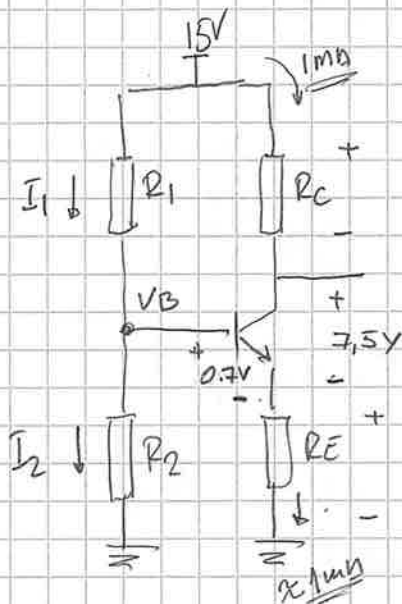
$$(1) v_{in}(t) \leq 6V$$

$$(2) v_{in}(t) \geq -9V$$



BJT DESIGN

Bias current

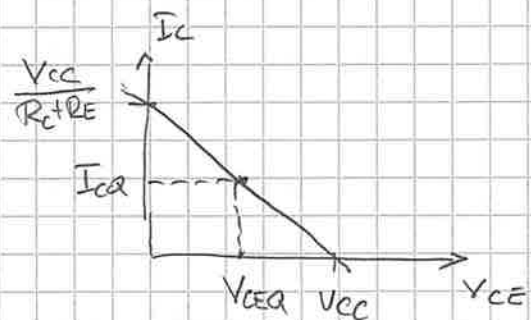


$$\frac{7.5}{4} \times 3 = 5.6V$$

$$R_C = \frac{5.6V}{1mA} = 5.6k\Omega$$

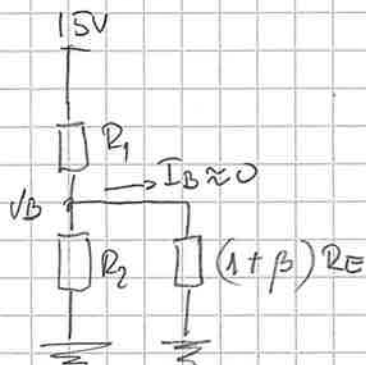
$$\frac{7.5}{4} = 1.9 \approx 2V$$

$$R_E = \frac{2V}{1mA} = 2k\Omega \approx 2.2k\Omega \quad (\Rightarrow V_E = 2.2V)$$



$$V_B = V_E + 0.7V = 2.2V + 0.7V = 2.9V$$

$$R_1 = \frac{15V - 2.9V}{I_1} = \frac{12.1V}{290\mu A} = 42k\Omega$$

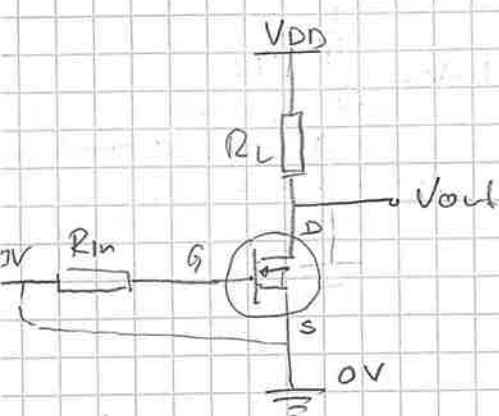


$$I_1 R_2 \ll \beta R_E = 100 R_E$$

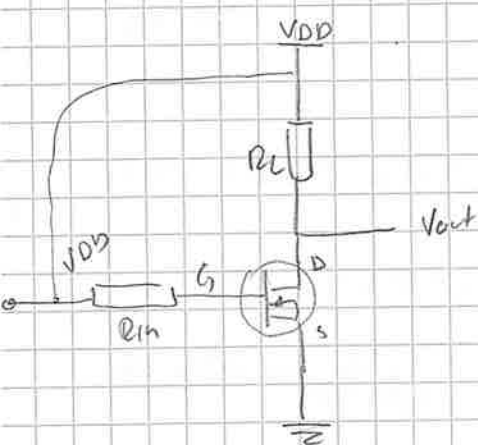
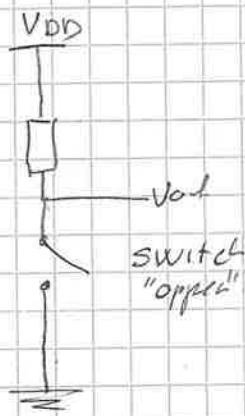
$$R_2 \ll 10 R_E \Rightarrow R_2 = 10k\Omega$$

$$I_2 = \frac{V_B}{R_2} = \frac{2.9V}{10k\Omega} = 290\mu A \approx I_1$$

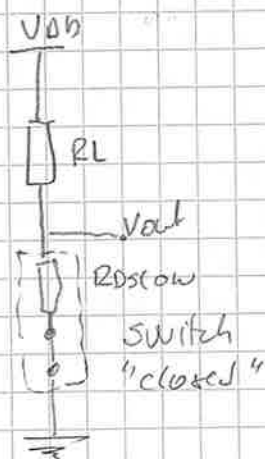
MOSFET AS SWITCH



- Input and Gate are grounded (0V)
- $V_{GS} < V_{TH}$
- MOSFET "OFF" (Cut-off region)
- $I_D = 0 \text{ A}$
- $V_{out} = V_{DS} = V_{DD} = "1"$
- MOSFET operates as open switch

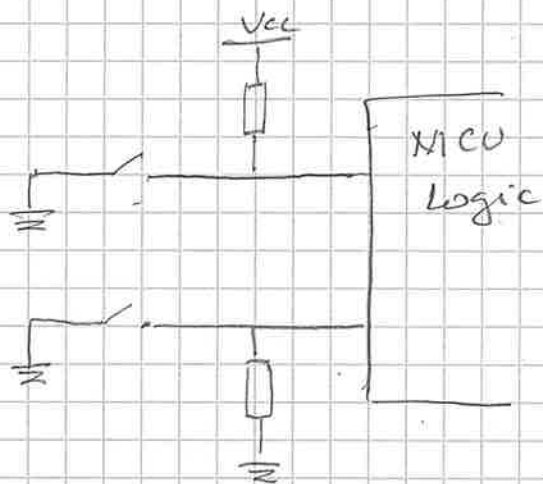


- Input and gate is connected to VDD
- $V_{GS} > V_{TH}$
- MOSFET is "ON" (saturation region)
- $I_D = \frac{V_{DD}}{R_L}$
- $V_{DS} = 0$ (ideal saturation)
- $R_{DS(on)} < 0.1 \Omega$ (minimum)



- $V_{out} = V_{DS} \approx 0.2 \text{ V}$ due to $R_{DS(on)}$
- MOSFET operates as Low resistance "Closed Switch"

Pull-up / pull down Resistors



7400, 74244

$$V_{IH} = 2V \sim V_{CC}$$

$$V_{IL} = 0.8V \sim 0V$$

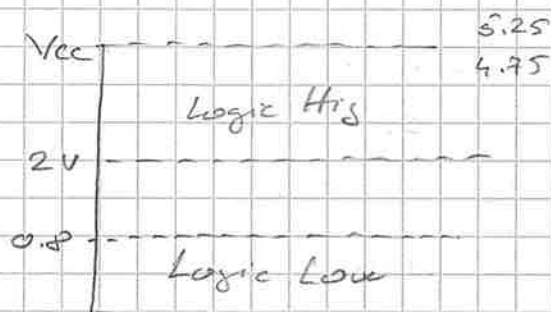
$$I_{IH} = 20 \mu A$$

$$I_{LH} = -400 \mu A$$

Pull up:

$$R = \frac{V_{CC} - V_{IH}}{I_{IH}}$$

$$= \frac{5 - 2}{20 \cdot 10^{-6}} = 150k$$

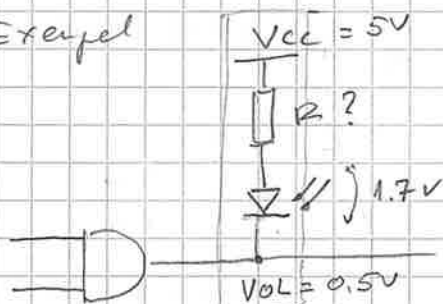


Pull-down:

$$R = \frac{V_{IL(max)} - 0V}{I_{IL}}$$

$$= \frac{0.8 - 0}{-400 \cdot 10^{-6}} = 2k\Omega$$

Example



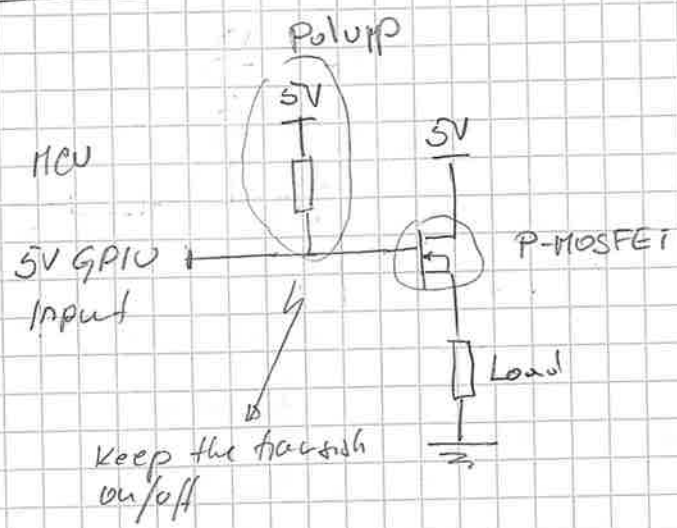
$$R = \frac{V_{CC} - V_{LED} - V_{OL}}{I_{LED}} = \frac{5 - 1.7 - 0.5}{15 \cdot 10^{-3}} = 186$$

$$\rightarrow V_{CC} - R I_{LED} - V_{LED} - V_{OL} = 0$$

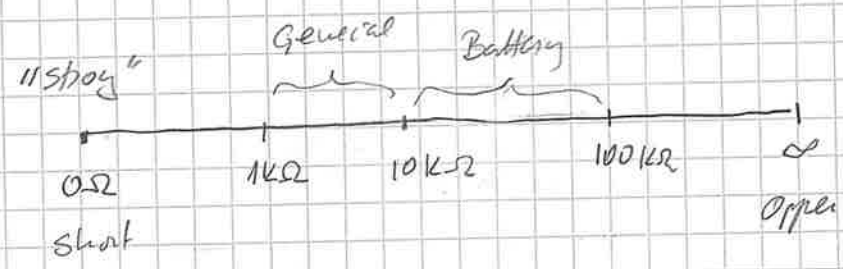
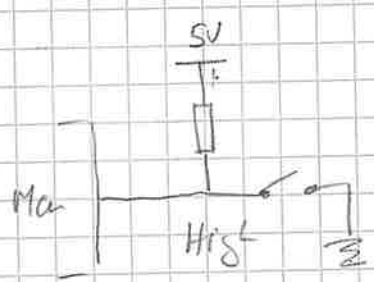
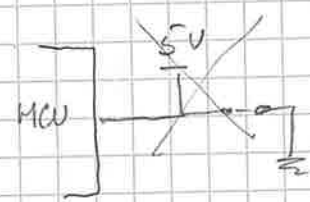
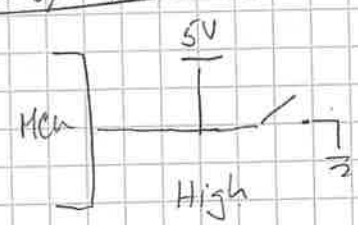
$$\Rightarrow$$

$$R = \frac{V_{CC} - V_{LED} - V_{OL}}{I_{LED}}$$

P-MOSFET

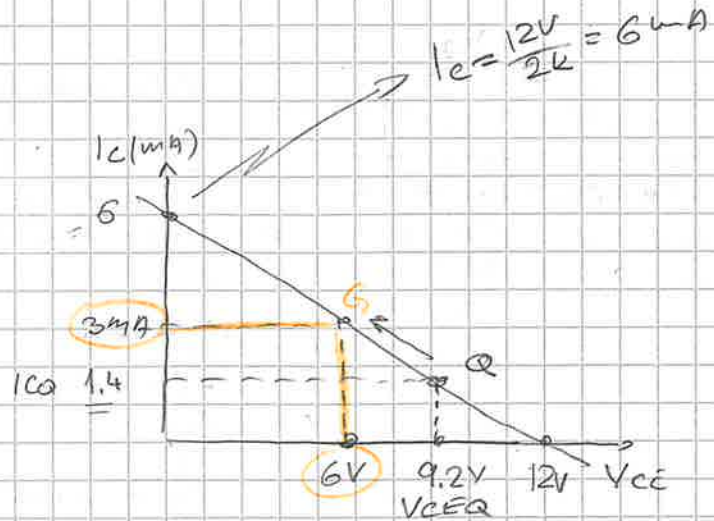
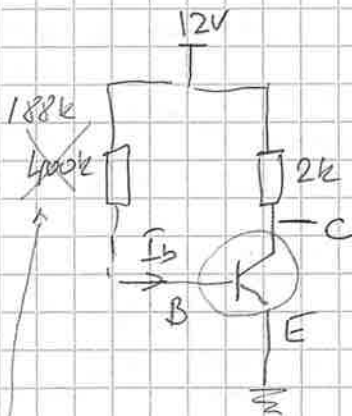


Pull up resistors



TRANSISTOR ($\beta = 50$)

Load line exaplo



if $I_{CQ} = 3mA$

$I_{BQ} = \frac{3mA}{50} = 60\mu A$

$R_D = \frac{12 - 0.7}{60\mu A} = 188.1k\Omega$

$I_B = \frac{12 - 0.7}{188k} = 60.1\mu A$

$I_C = 50 \cdot I_B = 3mA$

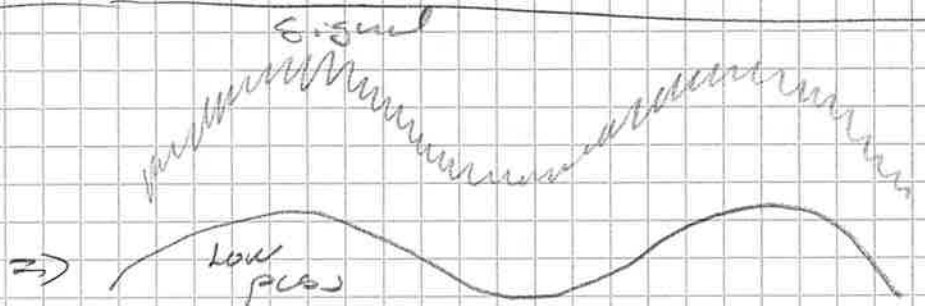
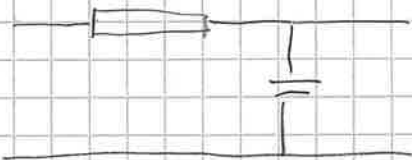
$V_{CE} = 12 - 2k \cdot I_C = 6V$

$I_B = \frac{12 - 0.7}{400k} = 28.25\mu A$

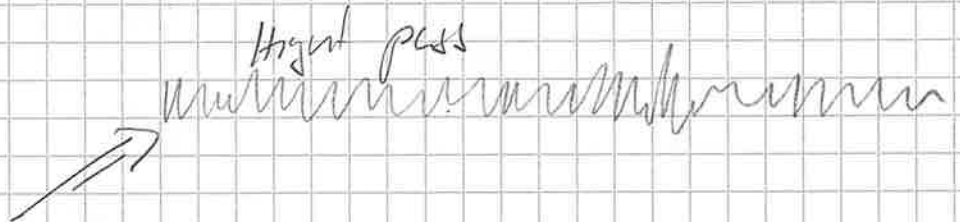
$I_C = 50 I_B = 1.413mA$

$V_{CE} = 12 - 2k \cdot I_C = 9.175V > 0.2V$

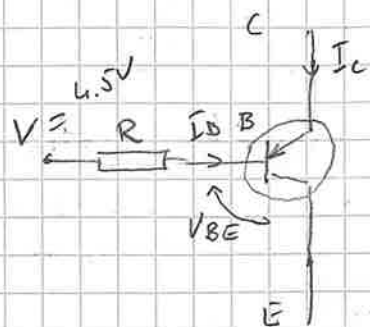
Low pass filter



High pass filter



Calculate base resistor for transistor



Ass switch I_B is $1/10$ of I_C

$I_B = 4 \mu A$ or $I_C = 40 \mu A$ ^{datasheet} max

$$R = \frac{V - V_{BE}}{I_B} = \frac{4.5 - 0.7}{0.004} = 950 \Omega \text{ or } 1k\Omega$$

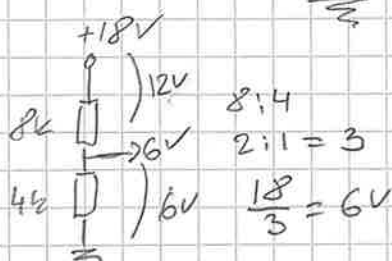
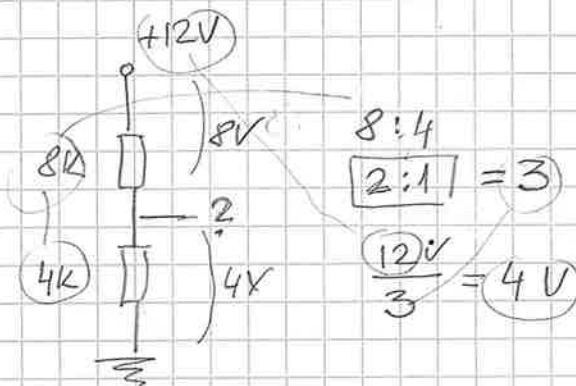
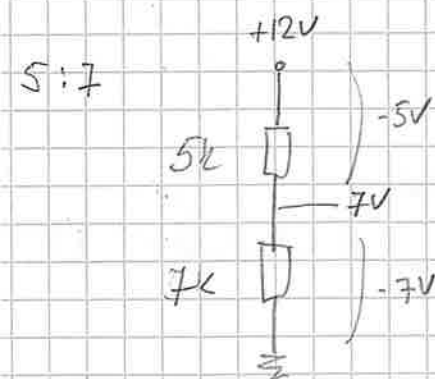
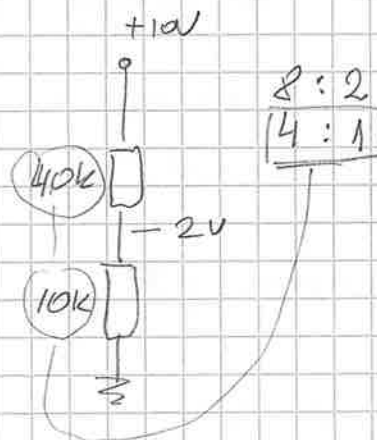
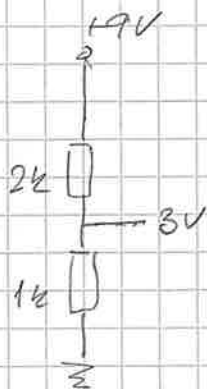
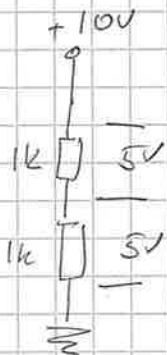
$H_{FE} = 100$ (use alt. min)

$$\frac{I_C}{I_B} = 100 \Rightarrow I_B = \frac{I_C}{100} = \frac{40 \cdot 10^{-3}}{100}$$

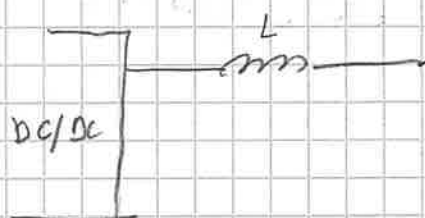
$$I_B = 40 \cdot 10^{-5} = 400 \mu A$$

$$\Rightarrow R = \frac{4.5 - 0.2}{400 \cdot 10^{-6}} \approx 9.5 k\Omega$$

Voltage divider



Inductance (DC/DC)



Inductance range 4.7 μ H - 10 mH

- Value chosen based on desired current ripple

- ripple current: from 20% of the average inductor current

$$Q = \frac{X_L}{R_s} = \frac{\omega L}{R_s} = \frac{\text{Reactance}}{\text{Total Resistance}}$$

= smaller loss $\Rightarrow$ high Q

- High V_{in} or V_{out}

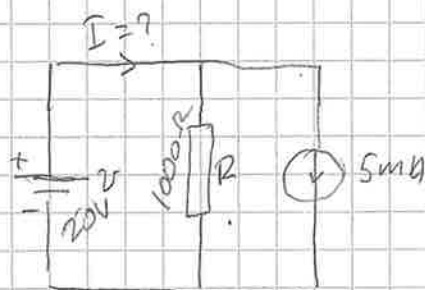
$\Rightarrow$ increase ripple current

Loss of Inductor

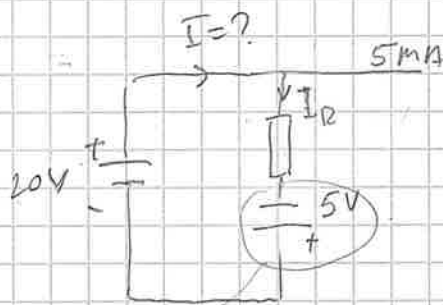
$$\Delta I_L = \frac{1}{f \cdot L} V_{out} \left(1 - \frac{V_{out}}{V_{in}}\right)$$

High V_{out} ripple } $\Rightarrow$ higher output current slew rate
Small-value L

L large $\Rightarrow$ lower the ripple current



Realni struj

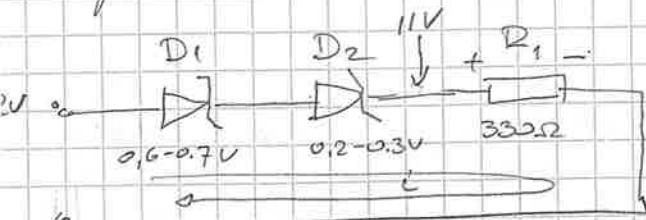


$$\Rightarrow \bar{I} = \frac{20V + 5V}{1000\Omega} = 25mA$$

$$I = 25 - 5 = 20mA$$

Realni naponski $U = I \cdot R = 1000\Omega \cdot 5mA = 5V$

Manipulation vid diode

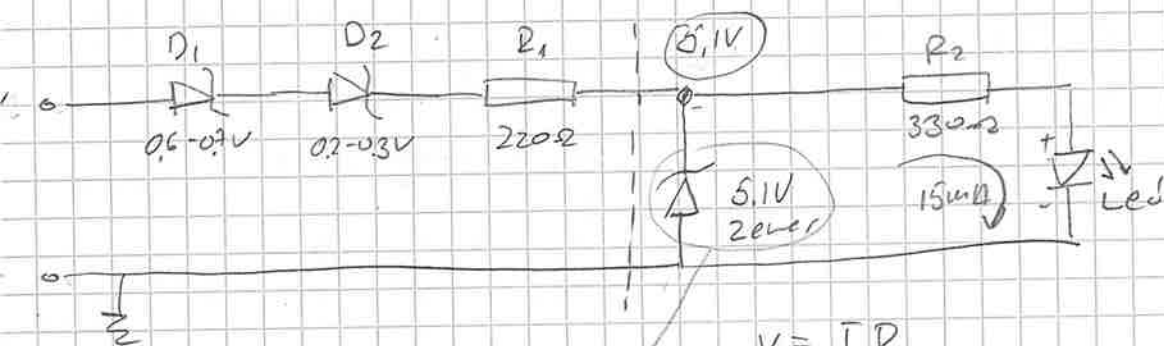


$$V = I \cdot R$$

$$I = \frac{11V}{330\Omega} = 33mA$$

$$V_{R1} = 12 - \underbrace{0.7}_{D1} - \underbrace{0.3}_{D2}$$

$$V_{R1} = 11V$$

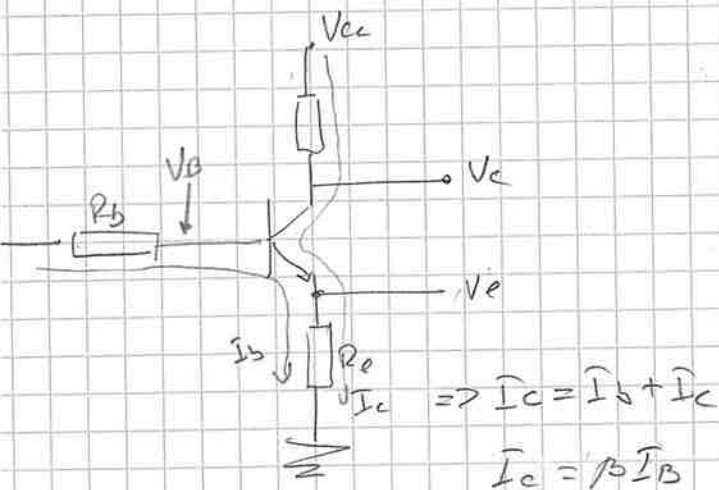


$$V = I R$$

$$I = \frac{5.1V}{330\Omega} = 15mA$$

Zener diode
is as
regulator

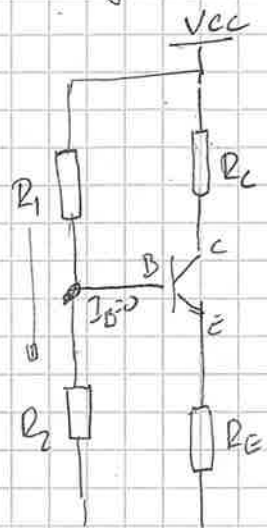
Transistor



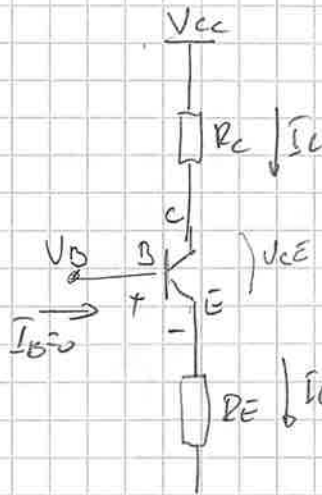
$$\Rightarrow I_c = I_b + I_e$$

$$I_c = \beta I_b$$

Voltage Divider Bias Configuration



$$V_B = \frac{R_2}{R_1 + R_2} V_{CC}$$



$$I_B = 0$$

$$I_C \approx I_E$$

$$V_E = V_B - V_{BE}$$

$$I_E = \frac{V_E}{R_E}$$

$$V_{CC} - I_C R_C - V_{CE} - V_E = 0$$

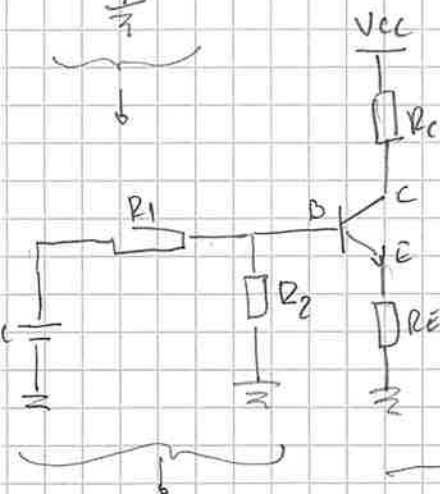
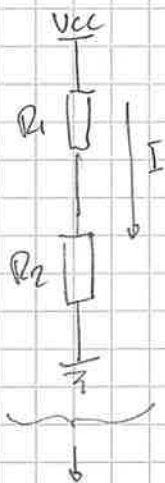
$$V_{CE} = V_{CC} - I_C R_C - I_E R_E$$

$$V_{CE} = V_{CC} - I_C R_C$$

$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$

$$V_E = V_B - V_{BE}$$

$$I_E = \frac{V_E}{R_E}$$



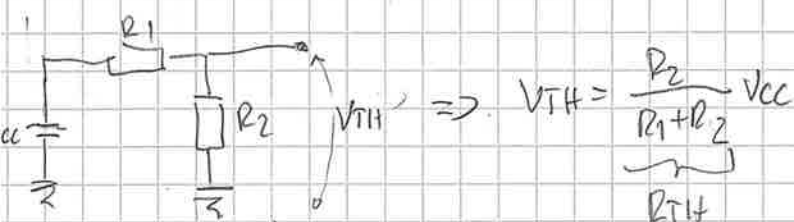
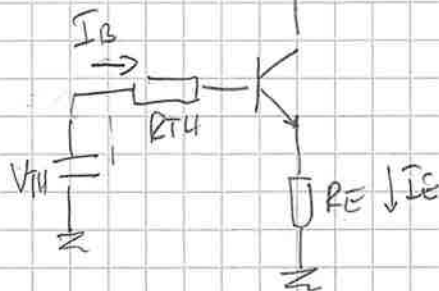
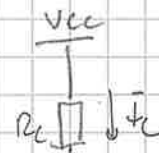
$$I_E = (\beta + 1) I_B$$

$$V_{TH} - I_B R_{TH} - V_{BE} - V_E = 0$$

$$V_{TH} - V_{BE} = I_B R_{TH} + I_E R_E$$

$$V_{TH} - V_{BE} = I_B R_{TH} + (\beta + 1) I_B R_E$$

$$I_B = \frac{V_{TH} - V_{BE}}{R_{TH} + (\beta + 1) R_E}$$



$$V_{TH} = \frac{R_2}{R_1 + R_2} V_{CC}$$

Voltage Divider Bias Configs

$$R_{TH} \ll (\beta + 1) R_E$$

$$R_{TH} \leq 0.01 (\beta + 1) R_E$$

$$\frac{R_{TH}}{\beta + 1} \ll R_E$$